Money and Power*

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Abstract

We link confidential loan-level regulatory data to power plant-level generation capacity and use two natural experiments to study how financial frictions affect electric power investment. We first show that environmental regulation raising the cost of operating coal power plants *tightened* financial constraints for exposed firms, which led to a contraction in the quantity and terms of credit and reduced their solar power investments. We then show that easing tax credit transferability for small firms *loosened* their financial constraints, which lowered their borrowing costs and boosted their solar power investments. Our results show that capital markets matter for energy policy.

Keywords: Electric power, financial frictions, energy policy

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1 Introduction

Demand for electric power in the US is surging at a pace not seen since the 1980s. After remaining flat from 2005 to 2020, total electricity supply has since been growing 2% per year (U.S. EIA, 2025), and this pace is expected to almost double over the next 15 years amid demand for energy-intensive facilities like data centers: Barth, Tai, Kaladiouk, and Heath (2025) project that demand could increase by up to 3.5% per year, while U.S. DOE, LBNL (2024) estimate that energy demand from data centers will increase from 4.4% of total US demand in 2023 to 12% by 2028. This demand cannot be satisfied with existing infrastructure and meeting it will require substantial additional investments in new generation capacity (U.S. Energy Information Administration, 2025a,c). Understanding these drivers of these investment decisions is thus of critical interest to policy makers.

Building the additional capacity necessary to meet this energy demand will incur massive up-front costs and take years to produce revenue. This timeline includes not just construction, which can take several years, but also time to obtain regulatory approval before construction starts and time to connect to the grid once it is complete (Johnston, Liu, and Yang, 2023). These large initial outlays and delayed returns make electric power generation a textbook example of an industry in which investments are likely to exhibit strong dependence on external finance (Fazzari, Hubbard, and Petersen, 1988). When investment is subject to financial frictions, the ultimate effects of any policy seeking to influence investment decisions depends not only on how it affects the expected net present value of an investment, but also on how it affects the terms and quantity of external financing necessary to fund that investment. Despite the theoretical importance of this financing channel for investments in electricity generation capacity, data constraints have limited the ability of past work to test and quantify this mechanism directly.

In this paper, we combine confidential loan-level regulatory data with plant-level generation capacity to study how financial frictions distort firms' solar power investments. We analyze the two most consequential US policies targeting electricity generation in the past two decades: the Mercury and Air Toxics Standards (MATS) in 2015 and the Inflation Reduction Act (IRA) in 2022. We focus on investments in new solar capacity because it was the single largest source of new capacity during the past five years—accounting for more than one-third of all new generation capacity during that time—and is projected to account for the majority of all new capacity growth between now and 2030 (U.S. Energy Information Administration, 2025b). While this makes solar the most relevant outcome for the policies we analyze, the mechanism underlying our results is not unique to solar, and we would expect any investments in new capacity to be distorted by financial frictions as long as they are expensive to build and slow to generate revenue.

We first show that MATS tightened firms' financial constraints and reduced their solar investments, even when the policy had, on paper, no statutory effect on solar power generation. MATS required roughly one half of coal-fired power plants to install "scrubbers," a costly pollution-control technology designed to reduce harmful emissions. Coal plants that failed to install scrubbers were required to shut down, which predictably led to decreases in coal generation capacity for firms exposed to MATS. However, many of these same firms also operated solar generation in addition to their coal plants, and it is a priori unclear how increasing the cost of coal energy production should impact investments in solar energy. We find that MATS exposure caused these firms to cut back on new solar investments. This response was driven by a tightening in financial frictions that propagated through these firms' internal capital markets: MATS exposure increased firms' reliance on asset-backed borrowing for their fossil lines of business, which allowed them to maintain the quantity and cost of credit for these operations, but caused their solar projects to receive fewer loans and at higher interest rates. These results illustrate how financial frictions caused a policy whose direct effects focused exclusively on coal power plants to reduce firms' solar energy investments.

Next, we leverage the IRA to show that easing financing frictions disproportionately stimulated solar investments for smaller firms with more binding liquidity constraints.

An important provision of the IRA allowed all firms to sell nonrefundable renewable energy investment tax credits to outside investors. This change did not matter for large firms with sufficient taxable income to claim the credits themselves, but it provided smaller firms with limited or no taxable income an additional source of up-front liquidity, thereby relaxing financial constraints. Consistent with this financing channel, we find that, after the IRA, smaller firms experienced larger reductions in interest rates on their solar loans and increased solar capacity investments relative to larger firms.

Together, these findings show how financial markets act as a key transmission mechanism for policies targeting electricity generation. The scale of investment necessary to meet growing demand for electricity in the US means that questions about what kind of new capacity will meet this demand, where it will get built, and who will operate it all have major implications for households, businesses, and government. Policy makers and regulators who seek to influence these investment decisions thus have a crucial interest in understanding what drives them and how they might respond to policy changes. While our results suggest failing to take these financial frictions into account can lead to unintended consequences—such as solar investment declining in response to restricting coal emissions—they also illustrate how the effect of subsidies can be amplified when they ease firms' financial constraints.

Related literature. Our paper contributes to two broad strands of literature. The first studies how taxes and subsidies designed to address externalities distort investment decisions in the presence of financial constraints and includes Tirole (2010), Hoffmann, Inderst, and Moslener (2017), Xu and Kim (2022), Martinsson, Sajtos, Strömberg, and Thomann (2024), and Döttling and Rola-Janicka (2025). We build on this work by providing direct empirical evidence for this channel in the context of electricity generation investments, which are subject to enormous economic and environmental externalities. The mechanism we study is similar to the ones described in Aghion, Bergeaud, De Ridder, and Van Reenen (2026), who use green patents as a motivating example for how financial constraints distort investments in new technologies more generally, and Lanteri and Rampini (2025), who analyze the role of financial constraints in determining the energy efficiency of firms' investments in the shipping industry.

The second strand of literature studies the determinants of firms' investments in energy generation capacity. Because energy is a critical input to virtually every part of the economy, electricity producers face a unique set of economic and regulatory incentives when making investment decisions. On this front, our paper is closely related to Darmouni, Lehner, and Sastry (2026), who use a theoretical framework to study the unique financing structure of renewable energy investments. Our paper complements their work by providing direct empirical evidence that these constraints matter for solar power investments using confidential loan-level data linked to plant-level generation activity. Our empirical evidence also builds on recent work including Johnston (2019), Hong, Kubik, and Shore (2023), Garrett and Shive (2025), and Sallee (2025) that analyzes how electric power generation has responded to a range of policy changes.

Other papers have focused on how taxes, subsidies, and research and development policies affect energy investment (Acemoglu, Aghion, Bursztyn, and Hemous (2012), Acemoglu, Akcigit, Hanley, and Kerr (2016), R.Brown, Martinsson, and Thomann (2022)). We provide empirical support for a "financial accelerator" mechanism, first described in Bernanke and Gertler (1989) and Kiyotaki and Moore (1997), in which the ultimate effects of any policy will depend on how it affects firms' financial constraints.

Our empirical evidence in the context of solar energy investment studies two policy changes: One which tightened financial constraints by raising costs and reducing revenues from non-solar operations (MATS), and one which eased financial constraints by providing additional up-front cash flows via tax credits (the IRA). The former finding is consistent with the mechanisms described in Lamont (1997) and Giroud and Mueller (2015), who show that firms' internal capital markets can propagate shocks across different lines of business in the presence of financial constraints. The latter finding supports Zwick and Mahon (2017), who show that bonus depreciation disproportionately boosted investment for small and financially constrained firms. We build on this work by highlighting the outsized importance of financial frictions in affecting investments in electricity generation capacity, whose investments are particularly sensitive to this channel.

This paper proceeds as follows. Section 2 describes our data and provides background about electric power financing. Section 3 shows that MATS reduced exposed firms' solar investments by tightening their financial constraints relative to unaffected firms. Section 4 shows that the IRA boosted investments for small and new firms by easing their financial constraints relative to larger and older firms. Section 5 concludes.

2 Data and Background

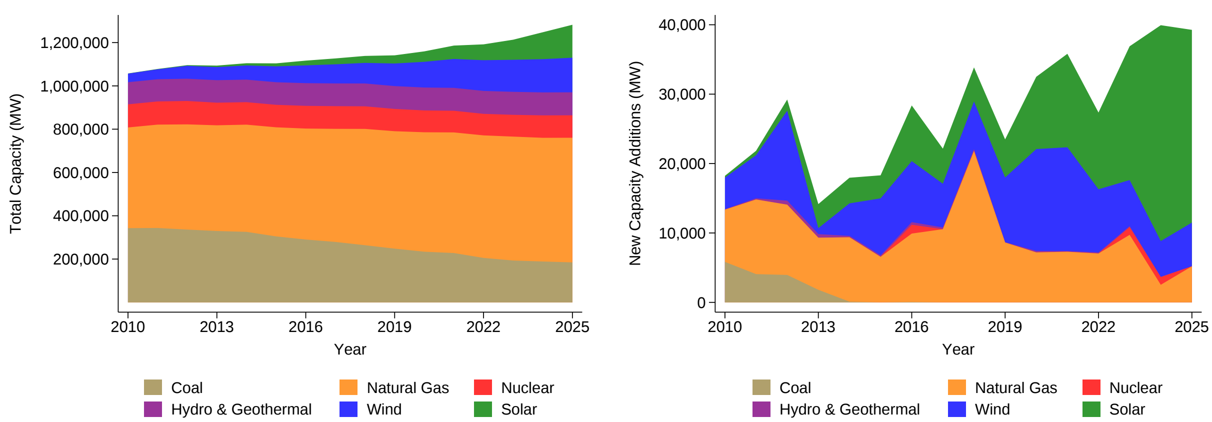
Here we provide background about electricity generation capacity in the US and describe how it is financed. We begin in Section 2.1 by showing the changing distribution of generation capacity by fuel type and emphasizing the important role of solar capacity investments. Next, in Section 2.2, we describe our loan-level regulatory data and use it to shed light on the financing arrangements of energy producers, including small and privately held firms. In Section 2.3, we discuss several key aspects of solar financing, including the use of bankruptcy-remote special purpose vehicles ("project finance"). Finally, in Section 2.4, we summarize these findings and discuss their implications for our subsequent empirical analysis.

2.1 Electricity Generation in the US

Data on electric power generation come from Form EIA-860 data from the US Energy Information Administration (EIA), which records annual information at the power plant generator level about generation capacity, fuel type, year constructed, utility name, and information about environmental control technology such as scrubbers. These data are well-suited for our purposes for two reasons. First, they provide utility names that allow us to link specific plants to borrowers in our financial data. And second, they provide detailed plant-level characteristics about both the stock and flow of different types of generation capacity over time; having this level of detail broken out specifically for new capacity being brought online is critical for understanding how the growing demand for power will be met going forward.

Figure 1 below illustrates the important distinction between the stock and flow of generation capacity in the US. The left panel shows that in 2010, roughly 80 percent of electricity generation capacity came from coal and natural gas. Since then, coal capacity has been declining steadily, with no meaningful new additions since 2014. In contrast, solar capacity has been rising steadily, as shown on the right panel of Figure 1. As of 2025, solar capacity accounted for roughly three quarters of total new capacity, and this trend is expected to continue going forward: According to the US Energy Information Administration's 2026 annual energy outlook, the majority of new capacity added between 2026 and 2030 is expected to be solar (Energy Information Administration, 2026).

Figure 1: Total (left) and new (right) generation capacity



Note: The left panel shows total electric power generation capacity in the US. The right panel shows new generation capacity. Source: EIA Forms 860 and 860m.

Hence, even though solar power still makes up a modest share of the stock of the total

US generation portfolio, it is still expected to be the single most important source of new capacity, which gives policy makers and regulators strong incentives to understand how solar investments are financed. We describe the details of these solar power financing arrangements and describe how they differ from other types of power generation in the next section.

2.2 Financial Data

Our primary source of borrowers' financial information is Schedule H.1 of the Federal Reserve's Y-14Q regulatory data. These filings, which are mandatory for all loans of $1 million or more held on the balance sheets of large[1] banks and bank holding companies, are used for stress testing as part of the Comprehensive Capital and Analysis Review (CCAR) program. Collectively, these banks hold more than 85% of total US banking sector assets and represent more than two thirds of all commercial and industrial (C&I) loan volume. The data, which are reported at the loan level every quarter, contain detailed loan characteristics, including each loan's size, interest rate, maturity, and what kind of collateral backs it. They report the borrower's taxpayer identification number (TIN/EIN), which we use as a common identifier to aggregate across different loans and merge in external data sources. They also include detailed income and balance sheet information and characteristics like name, industry, and location, and borrower risk assessments including perceived probabilities of default.

We focus on loans with reported[2] six-digit NAICS codes within electric power generation (221111 through 221118).[3] The left panel of Figure 2 plots total loan volume for different types of electric power generation during our sample. The patterns in the stock

[1]The threshold was $50bn in total assets when data collection began and increased to $100bn in 2019.

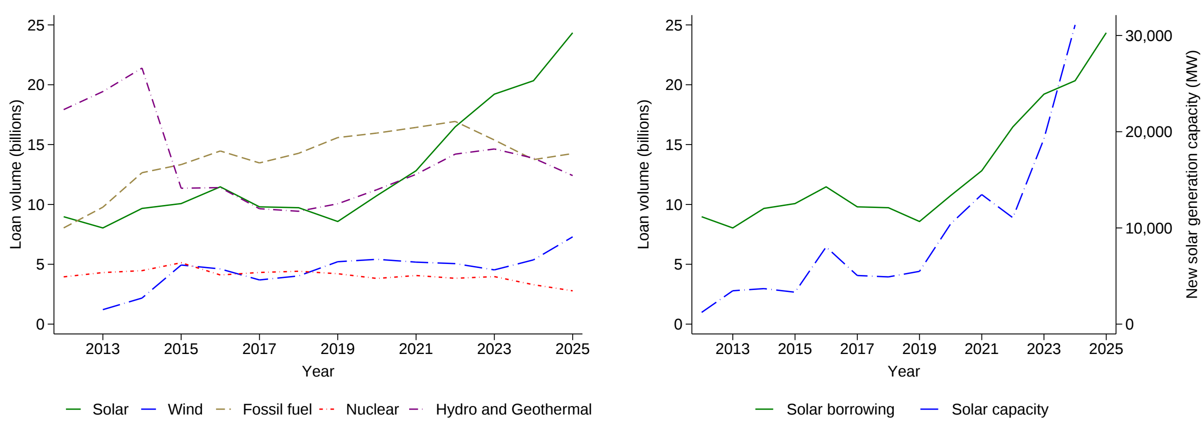
[2]Banks are instructed to "Report the numeric code that best describes the primary business activity of the obligor... If the business or individual operates in multiple industries, the [bank] should report the industry that best represents the commercial risk of the loan (i.e., the predominant industry)." See: https://www.federalreserve.gov/apps/reportingforms/Report/Index/FR_Y-14Q.

[3]The "fossil fuel" category includes coal and natural gas generation, both of which have the NAICS code 221112 and thus cannot be distinguished in our data.

of outstanding loans more closely mirrors the right panel of Figure 1 showing *new* capacity than the left panel showing total capacity. Indeed, the combined volume of renewable generation loans has surpassed fossil fuel every year in our sample, despite the fact that solar comprises a much smaller share of total generation capacity.

The right panel of Figure 2 compares these trends by plotting total solar loan volume against total new solar generation capacity by year. Solar loans and new solar capacity both started rising steadily in 2019, and in both cases had surpassed their counterpart values for fossil fuel in 2023. This is consistent with the idea that building new generation capacity is more capital intensive than simply operating existing capacity, and further illustrates why financial frictions are likely to be particularly important drivers of new capacity investments.

Figure 2: Electric generation loan volume by category



Note: This left panel shows total loan volume for the main electric power generation categories. Fossil fuel electric power generation includes both coal and natural gas generation (NAICS 221112). Data omitted for points with fewer than 20 loan observations. The right panel shows solar total loan volume (green, left axis) and annual solar capacity additions (blue, right axis). Source: Y-14Q and EIA form 860.

The Y-14Q data offer valuable insight into how firms finance these investments in new capacity because, in addition to providing detailed information about the specific terms of borrowers' bank loans, the data also report balance sheet measures that include loans from banks that fall outside of the scope of Y-14Q reporting and non-bank sources of financing. In the top panel of Table 1 below, we report summary statistics calculated from the solar loans in our sample. Because our unit of observation is a loan, we can only classify a borrower as a "solar firm" if we observe it receiving a solar power loan. Hence, all summary statistics for solar firms are based on the obligor characteristics reported for solar loans. If the same obligor TIN is reported on both a loan with a solar NAICS code and a loan with a fossil NAICS code, we include the characteristics reported on their solar loans to calculate summary statistics for solar borrowers, and the characteristics reported on their fossil loans to calculate summary statistics for fossil borrowers.[4] The median solar borrower has about $45 million of outstanding solar power loans in the Y-14Q data each year across three lenders at an interest rate of about 3.7 percent. Banks' median estimated probability of default for these borrowers is about 0.6 percent, and roughly 50 percent of loans are backed by specific collateral.

The median reported assets for an obligor receiving a solar loan is about $275 million, of which about two thirds are are fixed assets like land or equipment. The median borrower reports liabilities of approximately $150 million, of which more than three quarters comes from debt, suggesting that bank debt is a key source of financing for these producers. Indeed, many are unlikely to have other options; only 14 percent of the solar firms in our sample are publicly traded, and when we merge bond issuance data from Mergent FISD, we find that 94 percent of solar borrowers in our sample never issue any bonds.[5] The importance of bank financing for solar borrowers suggests that solar generation investments are likely to be highly sensitive to changes in the terms or availability of bank credit, a finding consistent with industry reports such as Lazard (2025), which shows that higher debt costs have larger effects on the total cost of solar power than for other types of generation.[6]

[4]For our analysis in Section 3, after aggregating loans to the obligor level using TIN, we also use a more rigorous manual matching process based on utility name and ownership data to aggregate across borrowers with different TINs and more accurately identify firms who operate both solar and fossil generation capacity.

[5]As we show in Appendix B.1, the overwhelming majority of bond issuance for electric power utilities is reported under the NAICS code for transmission and distribution rather than generation.

[6]Firms may also obtain loans from banks or other financial institutions that fall outside of the scope of Y-14Q. While our data do not include any information about these loans directly, the $45 million of bank

Table 2 reports the same statistics for fossil borrowers. As with solar, we define "fossil borrower" characteristics based on the reported obligor information for fossil fuel generation loans. Relative to solar borrowers, fossil borrowers receive more loans, have relationships with more banks, pay lower interest rates, and have longer loan maturity. Fossil borrowers are able to obtain more loans on better terms in part because they have a larger stock of productive assets and thus generate more revenue: The median fossil borrower generates more than $850 million in revenues from operating $3.2 billion of fixed assets, while the median solar borrower generates about $17 million in sales from $186 million in fixed assets. Fossil borrowers also have lower perceived default risk despite being less likely to use collateral, in part because many fossil generators are owned by regulated utilities with guaranteed revenues. This combination of lower revenues and higher risk can also help explain why solar borrowers face greater barriers to obtaining bank credit relative to their fossil counterparts. While we focus on solar and fossil borrowing here because they are the main subjects of our empirical analysis, we include summary statistics for other types of power generation in Appendix C.1.

From the perspective of a bank, these results are unsurprising. Financing the operations of an existing fossil fuel power plant is a less risky proposition than financing a new solar farm that might not generate revenue for several years, so it is natural that banks would be willing to extend more credit and provide better terms to fossil borrowers. However, solar projects have one important advantage over their fossil counterparts: Once installed and operational, the costs of producing solar power are negligible. In the next section, we discuss how solar firms are able to credibly pledge the expected future cash flows from these projects to obtain more funding without tying up their own balance sheets.

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loans that we do observe from Y-14Q banks for the median solar borrower represents about 40 percent of their total debt.

Table 1: Summary statistics for Solar borrowers and loans in the Y-14Q data

	Mean	Median	5%	95%	SD	N
Panel A: Firm characteristics						
Total number of loans	1.91	1.00	1.00	5.00	1.80	2,722
New loan (indicator)	0.41	0.00	0.00	2.00	0.98	2,722
Number of banks	3.09	3.00	1.00	8.00	2.27	2,722
Total loan volume ($ mil)	117.68	45.39	2.39	474.31	224.71	2,722
Interest rate (pp)	3.72	3.71	1.04	6.17	1.67	2,030
Maturity (months)	78.67	70.50	9.00	181.00	56.49	2,707
PD (pp)	2.32	0.62	0.11	8.00	8.61	1,807
LGD (ratio)	0.35	0.35	0.08	0.55	0.15	1,805
Collateralized (indicator)	0.50	0.50	0.00	1.00	0.47	2,722
Project finance (indicator)	0.25	0.00	0.00	1.00	0.41	2,722
Assets ($ mil)	7,401.08	274.29	4.11	46,259.00	20,225.72	1,637
Fixed assets ($ mil)	4,223.60	186.73	1.49	25,548.61	11,039.12	1,549
Liabilities ($ mil)	5,166.04	148.75	1.90	32,483.00	14,293.16	1,631
Debt ($ mil)	2,503.11	114.79	1.29	16,937.80	6,469.06	1,515
Sales ($ mil)	1,549.15	17.16	0.00	10,647.21	4,291.18	1,952
Publicly traded (indicator)	0.14	0.00	0.00	1.00	0.32	2,722
Bond issuer (indicator)	0.06	0.00	0.00	1.00	0.24	2,722
Value of bonds issued ($ mil)	1,318.84	850.00	200.00	3,800.00	1,305.97	168
Panel B: Loan characteristics						
Loan size ($ mil)	26.51	14.21	1.32	83.71	50.06	5,195
Interest rate (pp)	3.69	3.57	1.13	6.19	2.02	3,173
Maturity (months)	77.19	69.00	8.67	181.00	56.26	5,157
PD (pp)	2.75	0.66	0.12	9.76	10.26	3,075
LGD (ratio)	0.34	0.33	0.10	0.53	0.15	3,068
Term loan (indicator)	0.31	0.00	0.00	1.00	0.46	5,195
Revolver (indicator)	0.34	0.00	0.00	1.00	0.47	5,195
Collateralized (indicator)	0.45	0.00	0.00	1.00	0.48	5,195
Project finance (indicator)	0.27	0.00	0.00	1.00	0.44	5,195
Assets ($ mil)	6,978.60	360.54	5.69	40,337.00	18,681.73	2,767
Fixed assets ($ mil)	4,175.43	293.31	1.94	22,452.67	10,705.33	2,649
Liabilities ($ mil)	4,952.84	220.94	2.38	28,329.46	13,356.88	2,760
Debt ($ mil)	2,359.80	157.85	1.48	15,716.00	6,140.89	2,553
Sales ($ mil)	1,420.28	18.92	0.00	10,344.00	3,973.32	3,555

Note: Panel A shows summary statistics for the borrowers in our sample calculated by aggregating information about all loans with a reported NAICS code 221114 (solar electricity generation) to the obligor-quarter level using TIN and then taking simple averages across all variables to obtain a borrower-year panel. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. Reported values of zero are excluded from calculations for assets, fixed assets, liabilities, and debt. All variables come from the Y-14Q except for "Bond issuer" and "Value of bonds issued", which come from Mergent FISD. Panel B shows summary statistics for all loans with a reported NAICS code 221114 (solar electricity generation). All variables are calculated as simple averages across all loans. In both panels, borrower-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. All variables are described in detail in Appendix A. Source: Y-14Q and Mergent FISD.

Table 2: Summary statistics for Fossil borrowers and loans in the Y-14Q data

	Mean	Median	5%	95%	SD	N
Panel A: Firm characteristics						
Total number of loans	2.27	2.00	1.00	6.00	1.90	2,029
New loan (indicator)	0.30	0.00	0.00	2.00	0.81	2,029
Number of banks	4.18	4.00	1.00	11.00	2.93	2,029
Total loan volume ($ mil)	209.83	100.00	5.00	745.94	356.80	2,029
Interest rate (pp)	3.19	2.99	1.26	5.86	1.43	1,160
Maturity (months)	81.54	71.99	27.70	158.41	47.09	1,983
PD (pp)	2.15	0.33	0.05	8.00	8.15	1,479
LGD (ratio)	0.34	0.36	0.12	0.53	0.13	1,472
Collateralized (indicator)	0.29	0.00	0.00	1.00	0.42	2,029
Project finance (indicator)	0.14	0.00	0.00	1.00	0.32	2,029
Assets ($ mil)	14,765.18	3,740.05	119.64	69,996.75	26,235.35	1,798
Fixed assets ($ mil)	8,941.09	2,677.28	46.55	39,649.86	14,752.31	1,770
Liabilities ($ mil)	10,221.31	2,636.51	43.11	46,430.00	18,098.74	1,789
Debt ($ mil)	5,290.93	1,481.09	40.93	26,909.40	9,115.83	1,742
Sales ($ mil)	3,287.21	851.60	0.00	16,180.77	5,468.74	1,913
Publicly traded (indicator)	0.35	0.00	0.00	1.00	0.45	2,029
Bond issuer (indicator)	0.16	0.00	0.00	1.00	0.37	2,029
Value of bonds issued ($ mil)	1,051.03	700.00	250.00	3,400.00	1,109.38	322
Panel B: Loan characteristics						
Loan size ($ mil)	33.56	22.00	1.95	111.11	41.50	4,603
Interest rate (pp)	3.18	3.00	1.29	5.75	1.40	1,969
Maturity (months)	80.38	72.00	23.00	160.00	44.33	4,468
PD (pp)	1.98	0.37	0.05	8.00	7.65	3,255
LGD (ratio)	0.34	0.36	0.13	0.54	0.13	3,234
Term loan (indicator)	0.11	0.00	0.00	1.00	0.31	4,603
Revolver (indicator)	0.65	1.00	0.00	1.00	0.47	4,603
Collateralized (indicator)	0.29	0.00	0.00	1.00	0.44	4,603
Project finance (indicator)	0.15	0.00	0.00	1.00	0.36	4,603
Assets ($ mil)	15,305.44	4,457.03	120.12	64,729.00	25,101.13	3,934
Fixed assets ($ mil)	10,185.15	3,178.13	48.12	45,186.00	15,810.35	3,886
Liabilities ($ mil)	10,940.48	3,126.75	49.37	46,783.00	17,893.20	3,921
Debt ($ mil)	5,655.56	1,742.00	50.72	28,211.00	9,279.54	3,799
Sales ($ mil)	3,597.84	995.00	0.00	16,453.00	5,663.82	4,298

Note: Panel A shows summary statistics for the borrowers in our sample calculated by aggregating information about all loans with a reported NAICS code 221112 (fossil electricity generation) to the obligor-quarter level using TIN and then taking simple averages across all variables to obtain a borrower-year panel. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. Reported values of zero are excluded from calculations for assets, fixed assets, liabilities, and debt. All variables come from the Y-14Q except for "Bond issuer" and "Value of bonds issued", which come from Mergent FISD. Panel B shows summary statistics for all loans with a reported NAICS code 221112 (fossil electricity generation). All variables are calculated as simple averages across all loans. In both panels, borrower-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. All variables are described in detail in Appendix A. Source: Y-14Q and Mergent FISD.

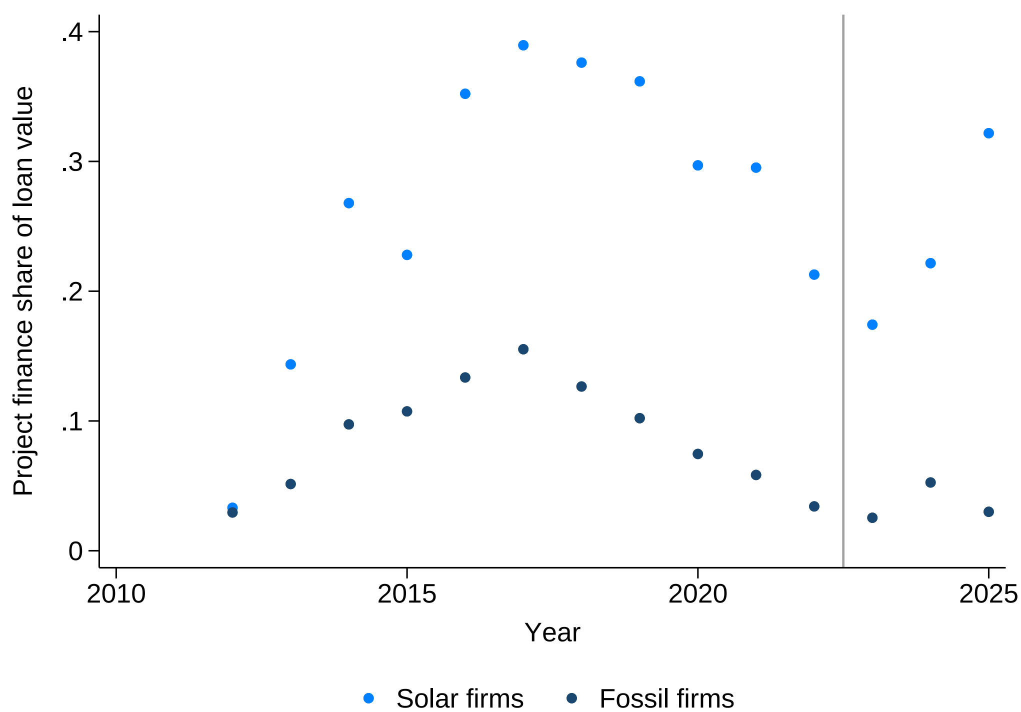
2.3 Solar Power and Project Finance

Many new solar generation projects in the US are financed using a non-recourse (or limited recourse) structure in which the loan is ultimately secured solely by the underlying solar assets and the cash flows they generate. This arrangement, which insulates the parent company's assets from creditors in the event the solar project fails, is commonly known as "project finance." Because installing new solar capacity has large upfront costs, these arrangements allow borrowers more flexibility in expanding capacity without severely constraining their balance sheet. Project finance helps mitigate some of the major risks associated with solar investments—including construction timeline risks, regulatory risks, difficulty interconnecting with the grid, and potentially unpredictable cash flows from generation (Jadidi, Firouzi, Rastegar, Zandi, and Eicker, 2025; Rand, Manderlink, Zhang, Talley, Gorman, Wiser, Seel, Kemp, Jeong, and Kahrl, 2025; Darmouni et al., 2026; Steffen, 2018)—while monetizing the reliability and low long-run operating costs from solar facilities as collateral for lenders.

Project finance generally consists of three components: (1) debt raised from banks or other lending institutions, (2) tax equity to monetize federal tax credits, and (3) sponsor equity. Debt financing is typically the largest financing source, making up around 50% to 60% of the project's total financing, and is usually secured from banks. Tax equity, which we describe in more detail in Section 4, makes up between 30% to 40% of the total financing, and can be obtained from banks or other large financial institutions. Finally, sponsor equity—which can come either from the "parent" firm whose assets are insulated from the special purpose vehicle or outside sources like hedge funds or private equity—make up the remaining 5% to 15% of a project's cost.

The debt used to fund the construction of new solar generation projects often comes from bank loans. These loans will show up in our data as long as they come from a bank subject to Y-14Q reporting and are at least $1 million. In the summary statistics tables below, we include a project finance indicator based on whether a loan either 1) reports a special purpose entity as the obligor, or 2) reports "project finance" as the loan's purpose.[7] Project finance loans account for roughly 25 percent of average solar loan volume for borrowers in our sample, compared to 14 percent for fossil borrowers. Figure 3 shows that this gap has widened over time, particularly following the passage of the Inflation Reduction Act (IRA), which is shown as the vertical line in 2022 and which provides identifying variation for our results in Section 4.

Figure 3: Project finance loan shares



Note: This figure shows the annual share of solar and fossil loan dollars classified as project finance loans. Observations before 2013 are omitted because of incomplete reporting in early years of the Y-14Q sample. The vertical line in 2022 corresponds to passage of the Inflation Reduction Act, which forms the basis of our identification strategy in Section 4. Source: Y-14Q.

In many cases, solar projects secure their debt financing with a power purchase agreement (PPA). A PPA is a contract in which a buyer agrees to buy a set amount of electricity for a set length of time at a given price. For example, a utility may contract to pay

[7]Because both of these definitions involve considerable discretion on the part of both the bank and the borrower, we view this measure as a useful but imperfect proxy for project finance as commonly understood by practitioners.

$35/mWh for 10 years for power from a solar facility. As Darmouni et al. (2026) emphasize, PPAs help solar projects obtain bank financing by providing predictable cash flows and mitigating the risks of fluctuating wholesale electricity prices. While we cannot explicitly observe when a borrower secures a loan with a PPA in our data, such a loan would count as "collateralized." Given that the special purpose entity will by construction have few other valuable assets, a bank's recourse for a project finance loan in bankruptcy will generally be limited to whatever revenues can be obtained through existing PPAs plus any remaining physical assets held by the project.

2.4 Summary

To summarize, this section showed that the borrowers of solar power loans exhibit more symptoms of financial constraint than their fossil counterparts: They are more reliant on bank debt, they receive smaller loans, and they pay higher interest rates despite being more likely to use collateral, all of which suggest that solar power investments—which are projected to account for the majority of new electricity generation capacity through the end of the decade—are likely to be highly sensitive to changes in the cost and availability of bank financing. We test these predictions in the data using identifying variation from an energy policy that tightened borrowers' financial constraints in Section 3, and an energy policy that loosened them in Section 4.

3 Evidence from MATS

This section analyzes firms' solar power investments responded to plausibly exogenous changes in environmental regulation that raised the cost of operating coal power plants. We begin by describing the policy, and its consequences for the sizeable share of solar power generation capacity operated by firms that also had coal power plants, in Section 3.1. In Section 3.2, we show that firms responded to these higher costs by cutting back not only on coal generation capacity, but also on solar generation capacity. In Section 3.3, we provide evidence that these spillovers are the result of the policy tightening exposed firms' financial constraints, highlighting the importance of firms' financial structure in driving their investment responses to policy changes.

3.1 Background

In 2012, the Environmental Protection Agency (EPA) first proposed the Mercury and Air Toxics Standards (MATS) to limit airborne emissions of mercury and other hazardous air pollutants created by coal power plants. To comply with MATS, all active coal-fired power plants were required to have or install "scrubbers," which clean the emissions from the power plant before they are released by exposing the polluted coal exhaust gasses to a liquid filtering agent. From 2012 to 2015, there was significant uncertainty as to whether MATS would ever be enforced: The Supreme Court had vacated a previous air quality regulation (the Clean Air Mercury Rule) in 2008, and legal cases were immediately filed against MATS by a number of states' attorneys general. Gowrisankaran, Langer, and Reguant (2024) estimate that the perceived enforcement probability fell as low as 43% in early 2015. In June 2015, the Supreme Court ruling *Michigan v EPA* did not overturn the MATS rule, effectively allowing it to take effect.

The penalties for noncompliance for MATS were large enough that all plants complied with the regulation by the April 2016. Table 3 reports how this compliance was achieved. In 2014, there was around 326,000 MW of active coal generation, with around 54% of that capacity already meeting the standards set forth in MATS (U.S. Energy Information Administration, 2016a). For coal-fired power plants that did not have scrubbers, the only options for achieving compliance by April 2016 were to either (i) retire the plant, (ii) convert the plant to burn natural gas, or (iii) install scrubbers. All three of these options incurred substantial costs.[8]

[8] For a typical 1,000 MW coal-fired power plant, retirement would cost around $117 million (Raimi, 2017),

Table 3 shows that about 81% of coal capacity achieved compliance by installing scrubbers.[9] Rather than paying the substantial costs of installing and operating scrubbers, around 13% of the coal capacity in operation at the end of 2014 retired by April 2016 to comply with MATS. Only 6% converted to burn natural gas.

Table 3: How coal plants respond to MATS

Action	Capacity (MW)
Already in compliance (MW)	175,377
Install scrubber (MW)	121,231
Converted to natural gas (MW)	9,526
Retired (MW)	19,698
Total coal (MW)	325,832

Note: The table shows how coal plants active in 2014 responded by the 2016 MATS compliance deadline. Source: EIA Form 860.

3.2 Effect of MATS on generation capacity

Having established that MATS acted as an unanticipated cost shock for firms with unscrubbed coal plants, we next show that it led to declines in both coal and solar generation capacity[10] for treated firms relative to untreated ones, and that financial frictions played a key role in generating these spillovers.

Our empirical strategy uses the EIA 860 data on energy generation capacity and compares otherwise-similar firms (defined by their utility identifier) that differed only in terms of their unscrubbed coal generation capacity (and thus their exposure to MATS) prior to the Supreme Court decision in 2015 Q3. We restrict the sample to firms that had

while the cost of converting a coal plant to a natural gas plant depends on what type of conversion (replacing coal with natural gas as heat source or installing a combined cycle plant) and the plant characteristics. Installing an activated carbon injection scrubber system was estimated to cost between $10 million and $30 million in 2007 dollars, while running and maintaining this system for a typical power plant would cost an additional $12–$28 million annually (U.S. Environmental Protection Agency, 2011; U.S. EIA, 2017)

[9] Scrubbers were installed at around the same rate for both regulated and deregulated coal fired power plants. Appendix B.3 provides a discussion of the role of regulation in scrubbers and MATS compliance.

[10] As mentioned previously, we focus on solar generation capacity, because it has been the single largest source of new generation capacity over the past several years and is expected to account for the majority of new capacity investments over the rest of the decade.

both fossil fuel[11] and solar generation assets at any point between 2010 and 2024. We then use the obligor names in the Y-14Q data to manually match loans to each utility identifier. Our identifying assumption is that the construction of electricity generation capacity for treated and untreated firms were on similar trajectories prior to the implementation of MATS and would have remained so in the absence of the regulation. Specifically, we estimate the following regression:

$$Y_{it} = \beta[\text{Treated X Post}]_{it} + \zeta_i + \delta_t + \epsilon_{it} \tag{1}$$

where Y_{it} is the energy generation capacity owned by firm i in year t, ζ_i are firm fixed effects, δ_t are time fixed effects, and ϵ_{it} is the error term. β is our coefficient of interest, which captures the effect of MATS exposure. We define a firm as "treated" if it had at least 250 MW of unscrubbed coal generation capacity at the time the Supreme Court decision was made in 2015Q3.[12] We cluster the standard errors at the firm level and weight observations by the firms' highest total annual generation capacity observed during our sample window.

Table 4 presents the results of equation 1. Column (1) shows that firms exposed to MATS reduced their coal capacity by more than 800MW following MATS. This effect is both statistically and economically significant, as it represents more than 25% of exposed firms' pre-MATS unscrubbed coal capacity. These magnitudes are consistent with the reductions in coal capacity from a combination of retirements and conversions documented in Table 3, and indicate that such reductions were likely driven by MATS exposure.

It is not surprising that a policy raising the cost of operating coal power plants reduced coal generation capacity. What is less obvious is how this policy should affect firms' in-

[11]Fossil fuel includes either coal or natural gas. While these categories are separately reported in the EIA data, coal and natural gas generation cannot be distinguished in the Y-14Q data.

[12]We use a threshold in levels instead of percent of generation because the cost of the policy depends on the total unscrubbed MW of coal, not its share of the firm's generation fleet. A 250 MW threshold is large enough to capture meaningful costs to the firm to retrofit, but not so large as to miss smaller plants (or boilers) that were affected by the policy. Appendix Tables A24 and A25 report similar results using smaller and larger thresholds.

Table 4: Changes in generation capacity for MATS-affected firms

	Coal (MW)	Solar (MW)	Natural Gas (MW)
	(1)	(2)	(3)
Treatment × Post	-807.366^{***}	-371.387^{*}	-15.501
	(225.896)	(188.107)	(429.589)
Firm count	119	119	119
Treated firm count	36	26	35
Observations	897	897	897
R-squared	0.97	0.68	0.99

Note: This table shows the effects of MATS on total generation capacity by generation type between 2015 and 2020. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: EIA Form 860.

vestments in *other* types of generation capacity. This response is determined by two opposing effects. First, by increasing the relative cost of coal generation, we would expect MATS to tilt the composition of firms' generation capacity toward non-coal sources, all else equal. However, as we showed in the previous section, MATS also tied up firms' financial resources by requiring some combination of costly retirements and the installation of new scrubbers while reducing revenues. Given the importance of external financing for investments in new capacity, these costs could instead push firms to *reduce* their investments in non-coal capacity.

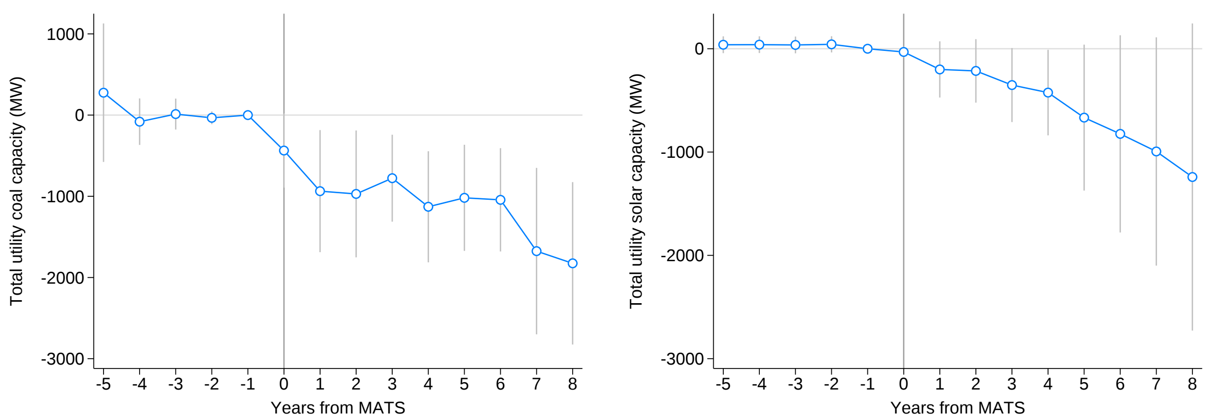
We find that the latter effect dominates. Column (2) shows large negative effects on firms' solar generation capacity. Solar capacity for treated firms expanded nearly 400MW less than untreated firms over the 5 years after MATS. This relative reduction in solar capacity is driven by stronger increases in solar construction for firms without MATS exposure. Column (3) shows that MATS did not cause a change in natural gas generation capacity for the firms in our sample.

Next, we explore the timing of the responses of firms' generation capacity to MATS. To the extent that firms responded by shutting down coal plants, these decisions should be reflected in an immediate reduction of coal generation capacity. On the other hand, investments in new generation capacity can take years to become operational, and we would therefore expect any effects on new solar investments to occur more gradually. To test these predictions formally, we estimate the following event study specification:

$$G_{it} = \sum_{k=-5,k\neq-1}^{8} \beta_k \cdot (\text{Treated}_i \times \mathbf{1}[\text{Years from MATS}]_{it} = k\}) + \zeta_i + \delta_t + \epsilon_{it} \qquad (2)$$

The results are shown below in Figure 4. The left panel shows that coal capacity declined sharply within a year of the Supreme Court decision that upheld MATS in 2015, suggesting firms' decisions to retire coal plants were both quickly implemented and permanent.[13] The right panel shows the relative decline in solar capacity among MATS-affected firms was more gradual, consistent with the treatment group building less new solar than the control group. As these investments gradually came online over the following years, the gap in solar capacity between treated and control firms steadily widened.

Figure 4: Change in coal capacity for MATS-affected utilities



Note: This figure shows the event study estimate from equation 2. The left and right panels show the cumulative effect of MATS on coal and solar generation capacity, respectively. Detailed variable definitions are in Appendix A. Error bars show 95% confidence intervals with standard errors clustered at the firm level. Source: EIA Form 860.

Financial frictions provide a natural explanation for why exposed firms fell behind their peers in solar investments even though MATS only targeted coal generation. By cutting cash flows while simultaneously requiring firms to pay for plant retirements or scrubbers, MATS weakened balance sheets, which could impact the cost and availability

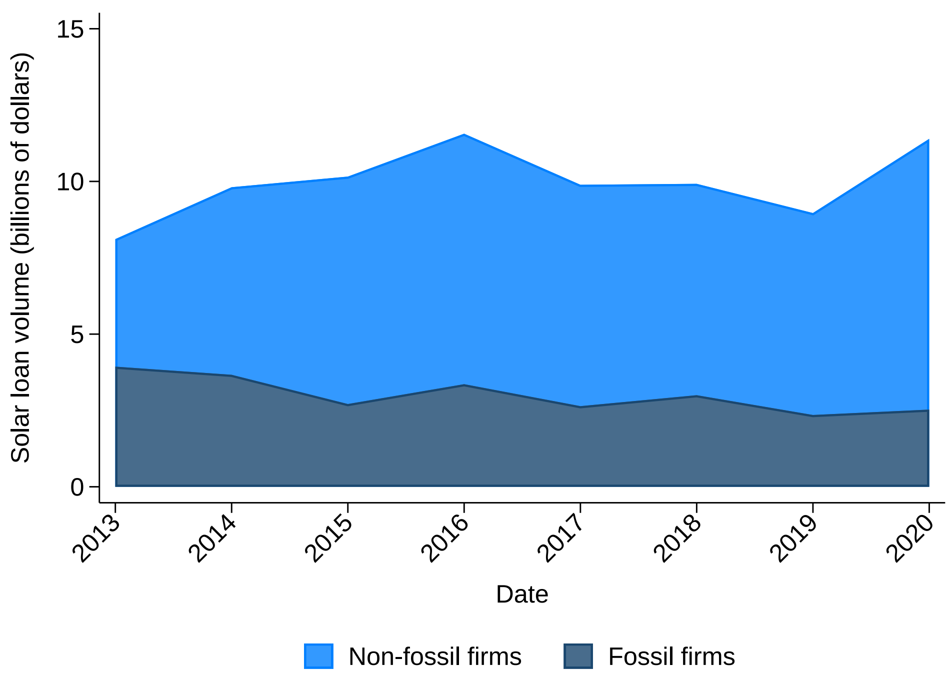
[13]We define retirement as no longer showing up as an active power plant in the EIA 860 data.

of external financing for both fossil and non-fossil investments. The next section leverages loan-level data to directly test these predictions.

3.3 Effect of MATS on financial outcomes

Here we show that financial spillovers can rationalize the effects of MATS on generation capacity documented in the previous section: While MATS exposure did not have any negative effects on the volume or terms of lending for exposed firms' fossil operations, it led to reductions in credit supply and higher interest rates for these same firms' solar operations. Figure 5 shows that the share of solar lending flowing to firms that also had fossil fuel generation loans in our data is nontrivial; in 2013 and 2014, around 40% of solar lending went to firms with fossil fuel loans, suggesting that shocks to fossil operations could have an economically significant impact on new solar investments.

Figure 5: Solar loans to fossil and non-fossil firms



Note: This figure shows the share of solar loan volume borrowed by firms that also borrow for fossil fuel electric generation. Source: Y-14Q.

Our empirical strategy compares firms that had unscrubbed coal generation capacity to otherwise-similar, but unaffected, firms. We use loan-level Y-14Q data, which contain detailed information about loan and borrower characteristics, but which cannot easily be associated with each plant's ultimate owner. To determine which firms were MATS-treated, we manually matched the Y-14Q data to the EIA 860 data using the firm name listed in both data sets.[14] Our set of manually-matched firms accounts for 62% of total lending volume for fossil fuel generation capacity in the Y-14Q, and 73% of the total fossil generation capacity in the EIA 860 data. To increase our sample size and power, we also include firms in the Y-14Q with solar and fossil loans that were not matched to the EIA data, either because their names did not line up with firm names in the EIA 860 data or because they did not own coal plants that we assume are not affected by MATS. Table 5 shows summary statistics for our Y-14Q variables before MATS went into effect.

Table 5: Summary statistics of firms in MATS sample

	Mean	SD	P10	P50	P90
Fossil committed exposure ($mil)	121.57	182.66	7.50	50.73	360.00
Fossil interest rate (%)	1.11	1.36	0.00	0.76	2.77
Fossil maturity (months)	48.41	28.91	16.00	45.95	82.39
Fossil percent of loans asset securitized	39.49	45.42	0.00	1.36	100.00
Fossil probability of default (%)	1.97	7.59	0.12	0.55	3.60
Solar committed exposure ($mil)	68.19	124.82	3.06	31.55	166.25
Solar interest rate (%)	1.81	1.79	0.00	1.65	3.91
Solar maturity (months)	62.62	45.57	9.67	51.12	128.66
Solar percent of loans asset securitized	52.49	47.36	0.00	61.73	100.00
Solar probability of default (%)	1.87	4.10	0.13	0.61	3.86

Note: This table show summary statistics for both fossil and solar borrowing before MATS went into effect (2013-2014). Detailed variable definitions are in Appendix A. Source: Y-14Q.

Given that the statutory effects of MATS were restricted to coal plants, we first analyze the response of firms' fossil loans by estimating equation 1 with financial outcome variables. The first column of Table 6 shows that MATS had no statistically significant

[14] We used a manual matching process, which we describe in detail in Appendix B.4, because there were many differences in firm names used in the Y-14Q and EIA 860 data. This approach provides a more comprehensive definition of a firm than our analysis in Section 2.2, which defined borrowers by solely by TIN, because some utilities borrow across multiple TINs.

effect on firms' total fossil loan volume. Next, we consider the effects of MATS on lending terms. Column (2) reports a tiny and statistically insignificant effect on interest rates of roughly one basis point despite a large (though not statistically significant) increase in exposed firms' perceived probability of default shown in Column (3).[15]

Table 6: Fossil loan outcomes

	Total loan volume (%)	Interest rate (pp)	Probability of default (pp)	Collateralized (pp)	Maturity (months)
	(1)	(2)	(3)	(4)	(5)
Treatment × Post	0.010	−0.192	4.421	15.985***	4.417
	(0.128)	(0.216)	(4.789)	(3.959)	(3.365)
Firm count	117	117	111	117	117
Observations	556	555	475	556	555
R-squared	0.89	0.77	0.58	0.87	0.83

Note: This table shows the effect of MATS on fossil loan outcomes. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q and EIA Form 860.

The combination of lower loan volumes and higher interest rates decreased despite greater perceived risk can be reconciled by Column (4), which shows that exposed firms increased their reliance on asset-backed borrowing as defined in Lian and Ma (2021). Past work such as Cerqueiro, Ongena, and Roszbach (2016) and Luck and Santos (2024) has shown that posting more collateral allows firms to obtain better loan terms, and we find evidence consistent with this channel: Following MATS, exposed firms increase their share of asset-backed borrowing by almost 16 percentage points, a 40% increase over their pre-MATS average rate of 39%. Finally, Column (5) shows that maturity increases by 4 months, representing a roughly 9% increase from its pre-MATS average. This result is consistent with Caglio, Darst, and Kalemli-Özcan (2024), who show in the Y-14Q data that firms tend to borrow at longer maturities when they are smaller, private, or more leveraged, all of which are traditionally associated with being more constrained.

Table 7 shows that MATS exposure did not have a statistically significant effect on the balance sheets of exposed firms' fossil operations. The point estimates for both assets and liabilities are positive, including estimated increases in fixed assets that would be consis-

[15] Probability of default is only reliably reported in the Y-14Q data starting in late 2014.

tent with the increased capital expenditures associated with installing scrubbers.[16] These estimates are noisy because our sample of fossil borrowers includes large utilities with diverse generation portfolios; while we can confidently assign the loan characteristics reported in Table 6 to fossil loans, we cannot definitively link changes in these borrowers' balance sheets to specific types of operations. Nonetheless, the point estimates are consistent with our earlier results in suggesting that exposed firms' fossil borrowing did not contract in response to MATS.

Table 7: Fossil borrower balance sheet outcomes

	Total debt (log)	Liabilities (log)	Assets (log)	Fixed assets (log)
	(1)	(2)	(3)	(4)
Treatment × Post	0.148	0.156	0.067	0.111
	(0.156)	(0.143)	(0.132)	(0.151)
Firm count	101	104	104	104
Observations	466	479	479	477
R-squared	0.90	0.94	0.95	0.96

Note: This table shows the effect of MATS on fossil loan outcomes. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q and EIA Form 860.

Together, these results suggest that firms exposed to MATS were able to avoid deterioration in the volume or terms of their fossil lending by increasing their use of collateral. This mechanism is consistent with Benmelech, Kumar, and Rajan (2024) and Rampini and Viswanathan (2025), who show that constrained firms rely more on asset-backed debt. Yet, as shown in section 3.2, the real effects of MATS were not confined to firms' fossil operations. If high compliance costs and increased use of collateral for fossil loans tightened exposed firms' financial constraints and limited their ability to obtain loans for their other lines of business, this could rationalize the large decline in solar investments documented above. Below, we provide empirical evidence consistent with this mechanism.

Table 8 reports the same outcomes as Table 6 for solar loans. Columns (1) and (2) show

[16] Many of the firms in our sample are rate-of-return regulated, and they can pass their costs along to their customers. However, that rate case regulatory process can take years, and in the case of MATS, financing was needed relatively quickly to install and run scrubbers.

statistically and economically significant declines in loan volume of 43% and increases in average interest rates of 78 basis points for exposed firms relative to their untreated counterparts. Column (3) shows a negative (though statistically insignificant) response in firms' probability of default, suggesting that increasing interest rates are not simply the result of exposed firms pursuing riskier solar projects. Column (4) reports that this worsening in solar loan terms occurred even though exposed firms increased the share of their solar loans that were explicitly collateralized by their assets. The estimate is imprecise, but the positive coefficient suggests that worsening solar terms are not a mechanical consequence of shifting collateral across lines of business. Instead, this result points to a broader deterioration of the firm's financial prospects that extended beyond fossil operations, even though MATS exclusively targeted coal plants.

Table 8: Solar loan outcomes

	Total loan volume (%)	Interest rate (pp)	Probability of default (pp)	Collateralized (pp)	Maturity (months)
	(1)	(2)	(3)	(4)	(5)
Treatment × Post	−0.434**	0.783**	−1.555	18.852	7.211
	(0.173)	(0.350)	(1.427)	(16.204)	(4.939)
Firm count	117	117	91	117	117
Observations	510	510	375	510	510
R-squared	0.90	0.62	0.66	0.83	0.85

Note: This table shows the effect of MATS on solar loan outcomes. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q and EIA Form 860.

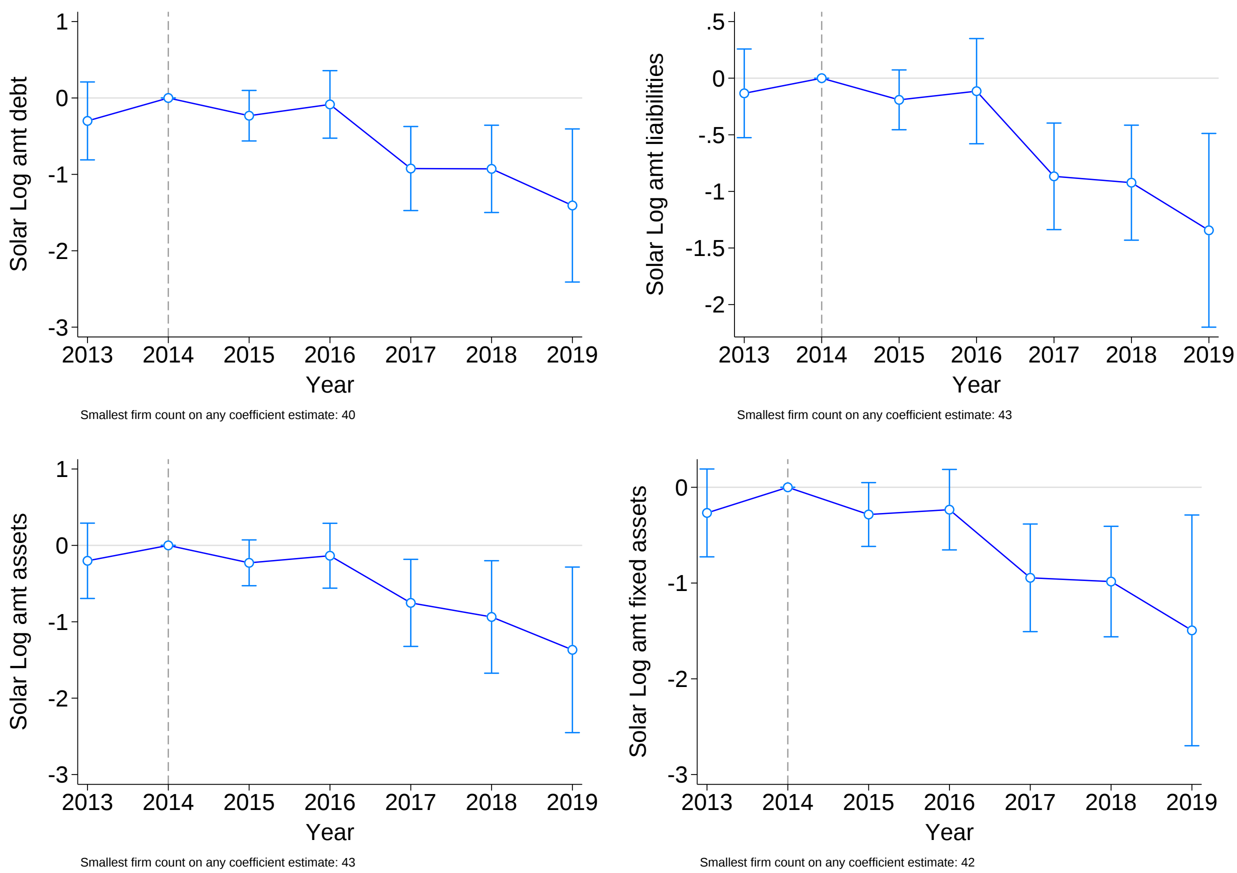
Table 9 shows that the balance sheets underpinning borrowers' solar loans shrank notably following MATS: The declines in total liabilities of 62 percent and total fixed assets of 65 percent are both economically large and statistically significant. We plot the corresponding event-study estimates in Figure 6 and suggest that MATS affected realized capacity with a four-to-six year lag; this aligns closely with both the estimated response of new capacity from the EIA data in Figure 4 and outside estimates of average build times (U.S. Energy Information Administration, 2016b; Nilson, Hoen, and Rand, 2024).

Table 9: Solar balance sheet outcomes

	Total debt (log)	Liabilities (log)	Assets (log)	Fixed assets (log)
	(1)	(2)	(3)	(4)
Treatment × Post	-0.547^{*}	-0.616^{**}	-0.571	-0.646^{**}
	(0.287)	(0.246)	(0.345)	(0.319)
Firm count	85	87	87	84
Observations	338	351	351	339
R-squared	0.93	0.92	0.94	0.95

Note: This table shows the effect of MATS on solar loan outcomes. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q and EIA Form 860.

Figure 6: Solar borrower balance sheet characteristics



Note: This figure shows the event study estimate from equation 2 using balance sheet characteristics reported by solar borrowers. The vertical dashed line corresponds to the last pre-treatement year. Detailed variable definitions are in Appendix A. Error bars show 95% confidence intervals with standard errors clustered at the firm level. Source: Y-14Q and EIA Form 860.

Taken together, this section showed that policies targeting electricity generation can have unintended and counterintuitive effects on firms' new capacity investments when those policies simultaneously tighten firms' financial frictions. While the direct compliance costs of MATS were entirely contained to firms' fossil operations, installing scrubbers and shutting down coal plants tied up financial resources that could otherwise have been put to use in other lines of business. By shifting resources away from solar operations—which, as we showed in Section 2.4, rely more on bank financing and tend to exhibit more symptoms of financial constraint—MATS reduced investments in solar capacity. This aligns closely with the mechanisms in Lamont (1997) and Giroud and Mueller (2015), who show that shocks can propagate through firms' internal capital markets and that the effects are strongest for industries whose investments are subject to greater degrees of financial constraint. Hence, firms' investment responses to policy changes will depend on how these policies affect the cost and availability of external financing. Having provided empirical evidence for this mechanism using a policy that tightened firms' financial constraints in this section, we next analyze a policy that relaxed them.

4 Evidence from IRA

In this section, we study the effects of the Inflation Reduction Act (IRA) on solar power investments. The IRA included several tax provisions that, while technically available to all firms, primarily benefited small and financially constrained producers. Following the IRA's passage, we find that these small producers disproportionately increased their investments in new capacity relative to larger producers. Consistent with the IRA easing financial constraints, we show that these smaller producers also experienced increases in loan volume and decreases in interest rates relative to larger firms.

4.1 IRA background

The most important provisions in the IRA concerned the transferability of tax credits.[17] The investment tax credit (ITC) and production tax credit (PTC), which provide subsidies for solar and wind development, have been available since 1978 and 1992, respectively. Before the passage of the IRA in 2022, both credits were non-refundable, meaning they could only be claimed by firms with tax liabilities. Because many small solar and wind developers do not have enough tax liability to obtain the full benefit from these credits themselves, they were historically forced to use complex financial arrangements called tax equity swaps to monetize these tax credits. Tax equity swaps, which we describe in more detail in Appendix B.2, pair a solar project with a tax equity investor—typically a large bank—who provides funding to the solar project in exchange for an equity stake in the project that allows them to claim the tax credits.[18]

However, these products were very expensive, resulting in developers without tax liabilities receiving only a portion of the full government subsidies: The nonrefundable PTC was valued at roughly 85 cents on the dollar (Johnston, 2019). Reliance on these costly derivatives meant that larger solar developers with tax liabilities benefited more from the ITC and PTC than smaller firms. The IRA allowed firms to sell the tax benefits directly to investors without having to set up a tax equity swap. At the time of the IRA's passing, public coverage of the provision highlighted how these new transferability rules were changing how solar and wind developers monetized the tax credits (U.S. Department of the Treasury, 2024; Rubin and Ramkumar, 2024; Weaver, 2024). While the transferability of tax credits technically applied to all firms, in practice, it effectively eased financial constraints for smaller firms relative to larger firms with tax liabilities. Hence, our empirical strategy examines the effects of the transferability of the ITC and PTC by comparing both real and financial outcomes for small firms to large firms before and after the IRA passed.

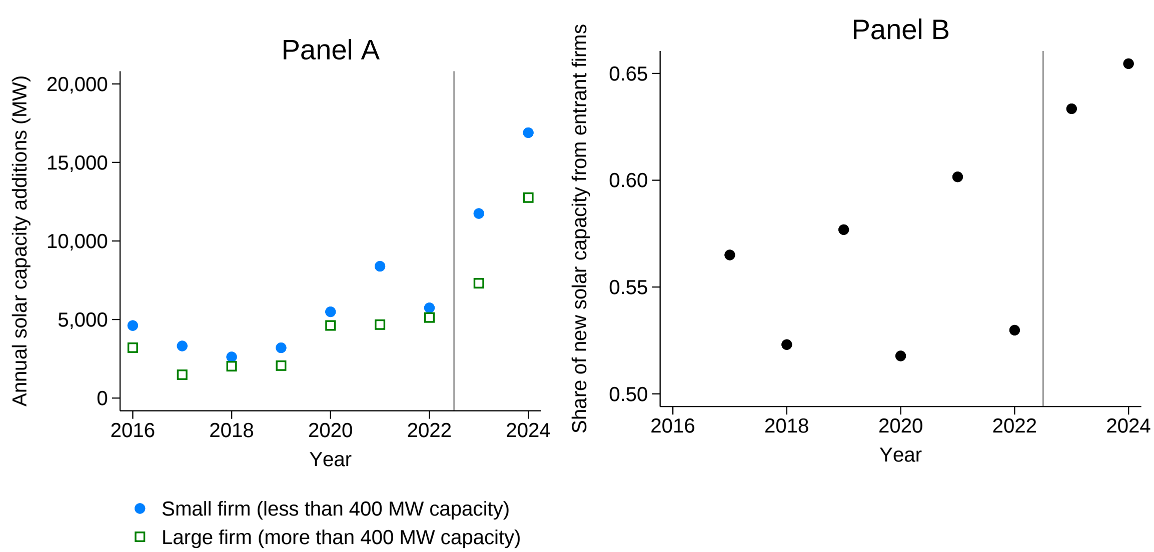
[17]See Aldy (2025) and Bistline and Wolfram (2025) for more details of the IRA and its other components.

[18]While Garrett and Shive (2025) show that tax equity investors can affect the operations of wind plants through monitoring and contracting, they find no effect on solar plants, which are the focus of our paper.

4.2 Effects of the IRA on generation capacity

We begin by providing suggestive evidence that the IRA resulted in more small producers entering the market to build solar capacity. Here we refer to "producers" instead of "firms" to make it clear that the results in this subsection come entirely from EIA data that do not include any of the firm-level identifiers in the Y-14Q. We define a producer by its EIA utility identifier and, after dropping producers with less than 5 MW of total generation capacity, classify a producer as "small" if its highest total generation capacity between 2016 and 2024 was less than 400 MW. Although this cutoff classifies roughly 95% of producers as small, we choose it for two reasons. First, it is unlikely that operating 400 MW of capacity during this time would generate sufficient tax liabilities to take advantage of the refundable tax credits that were made transferable under the IRA. And second, this cutoff also results in similar pre-IRA capacity additions for firms above and below the threshold. Figure 7 below compares new solar generation capacity built by these small producers to their larger counterparts in each year before and after the IRA.

Figure 7: New solar capacity added by firm size



Note: Panel A shows new solar capacity by year based on maximum firm size (in MW) over the sample. Panel B shows the share of new solar capacity that is being built by new entrants (firms that did not exist in the previous period). Source: EIA Form 860.

Prior to the IRA passing in 2022, large and small producers accounted for a similar total quantity of capacity additions each year. After the IRA was passed in 2022, however, smaller producers accounted for a notably larger share of new capacity additions.[19] The right panel shows a similar jump in the share of new solar capacity coming from new entrants—who overwhelmingly tend to be small producers—following the IRA's passage. While these results are not causal, they are consistent with the idea that the transferability of the ITC and PTC in IRA benefited smaller solar energy developers relatively more than larger ones. In the next section, we provide evidence suggesting these effects were a direct consequence of the fact that the IRA disproportionately eased financial constraints for smaller projects.

4.3 Effects of the IRA on bank lending

Given the Inflation Reduction Act's emphasis on easing financial constraints for smaller firms, our empirical strategy compares the effects of the IRA on financing conditions for small firms' solar energy projects relative to large ones. The larger pool of firm names for solar energy loans relative to fossil loans in the Y-14 renders the manual matching process used in Section 3.2 infeasible. Instead, for this exercise, we aggregate the loans in the Y-14Q data to firm-quarter panel using obligor TIN and define treated "small" firms as those with less than $30 million of outstanding electricity generation loans (including fossil, wind, solar, nuclear, and hydro power) in the Y-14 prior to the IRA, though our results are robust to a range of alternative thresholds. We estimate the following equation:

$$Y_{it} = \beta[\text{Small firm X Post}]_{it} + \zeta_i + \delta_t + \epsilon_{it} \tag{3}$$

Where Y_{it} is a bank finance outcome for firm i in year t, ζ_i is a firm fixed effect, δ_t is a time fixed effect, and ϵ_{it} is an error term, which we cluster at the firm level. *Post*

[19] Appendix Figure A6 finds similar results through 2025 using planned construction.

is an indicator for the passage of the IRA, which turns to 1 for all firms starting 2023. Our coefficient of interest is β, which captures the effect of the IRA on smaller firms solar borrowing behavior relative to larger firms. Our identification assumption is that the financing terms for solar energy projects were on similar trajectories for all firms prior to the IRA's unanticipated passing,[20] and that these similar trends would have continued in its absence. We only include firms that had solar loans in the year before the IRA passed to focus on the subset of firms best positioned to expand their solar capacity.

Table 10 shows the pre-period values of our outcomes of interest for smaller treated firms and larger control firms; summary statistics for the pooled sample are reported in Appendix Table A23. Treated firms have smaller loans, higher interest rates, and longer maturities than the larger control firms. These differences are consistent with larger firms being able to secure better interest rates and the findings in Caglio et al. (2024) that loans to firms generally considered to be less financially constrained tend to have slightly shorter maturities in the Y-14 data.

Table 10: Comparison of treated and control IRA firms in pre period

	IRA treated	IRA control	P-value of diff
Solar committed exposure ($mil)	10.02	84.50	0.000
Solar interest rate (%)	3.30	2.42	0.000
Solar maturity (months)	70.18	53.17	0.000
Solar percent of loans asset securitized	60.90	42.67	0.000
Solar probability of default (%)	2.60	2.58	0.952

Note: This table compares the treated and control group before the IRA went into effect (2019-2021). Source: Y-14Q and EIA Form 860.

Table 11 reports results of estimating equation 3. While Column (1) shows that the IRA's effects on loan volume were similar for all firms following the IRA, Column (2) shows that solar loans to smaller firms became much cheaper. The 65 basis point decline

[20] After extensive discussions throughout 2021 and early 2022, Congressional discussions led to an unexpected majority for the bill in late July 2022 https://www.politico.com/news/2022/07/27/manchin-schumer-senate-deal-energy-taxes-00048325, and the IRA quickly passed in August. The long deliberations and sudden bill approval suggest that the IRA was not anticipated by energy companies.

in the average interest rate on solar loans for small firms relative to large ones is statistically significant and economically large, representing almost the entire pre-period gap between IRA treated and control firms shown in Table 10.

Table 11: Solar loan outcomes

	Total loan volume (%)	Interest rate (pp)	Probability of default (pp)	Collateralized (pp)	Maturity (months)
	(1)	(2)	(3)	(4)	(5)
Treatment × Post	-0.055	-0.647^{***}	1.821	-0.448	-7.051^{**}
	(0.087)	(0.200)	(1.372)	(2.996)	(2.746)
Firm count	312	284	277	312	309
Treated firm count	162	140	134	162	159
Observations	1666	1467	1432	1666	1653
R-squared	0.91	0.85	0.60	0.89	0.91

Note: This table shows the effect of IRA on solar loan outcomes. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q and EIA Form 860.

How did the IRA reduce borrowing costs for small projects relative to large ones? One natural explanation is that, by reducing or removing the need for a costly tax equity swap allowed firms to fund the same project with lower leverage. Contributing more money to a project up front gives firms more "skin in the game," which mitigates the adverse consequences of asymmetric information for banks and expands the set of firms' potential investments.

Another non-mutually exclusive possibility is that the IRA induced a shift in the composition of firms' loans: Column (5) shows a statistically significant decline of about 6 months in the average solar loan maturity, representing roughly 10% of the pre-IRA average. Given that shorter-maturity loans tend to have lower interest rates, one possible explanation for the cheaper borrowing costs for small firms following the IRA is that these firms took advantage of their improved financial position to borrow at shorter maturities with lower term premia.

Because this analysis includes only firms that had solar loans prior to the IRA passing, it necessarily omits new entrants, who tend to be smaller and more financially constrained. Thus, we view these effects as a lower bound on the true magnitude of the degree to which the IRA eased small firms' financial constraints, and it is likely that the new entrants captured in the right panel of Figure 7 experienced larger declines in borrowing costs than our results here would imply. Consistent with this mechanism, Table 12 below shows that the reductions in interest rates and maturities following the passage of the IRA were larger for project finance loans. As we described in Section 2.4, project finance loans are structured to be isolated from the parent company's balance sheet, and thus far more likely to have low or no revenue at early stages. Hence, to the extent that the IRA eased firms' financial constraints by making it easier to monetize tax credits, it is unsurprising that these benefits accrued disproportionately to project finance loans.

Table 12: The effects of the IRA for small borrowers using project finance

	Total loan volume (%)	Interest rate (pp)	Probability of default (pp)	Collateralized (pp)	Maturity (months)
	(1)	(2)	(3)	(4)	(5)
Treatment × Post	−0.131	−0.504**	2.251	−1.090	−6.982**
	(0.087)	(0.208)	(1.545)	(3.181)	(3.052)
Project finance × Treatment × Post	0.099	−0.939***	0.078	0.841	−7.188**
	(0.122)	(0.338)	(1.193)	(3.135)	(3.576)
Firm count	312	284	277	312	309
Project finance firm count	120	112	93	120	120
Observations	1666	1467	1432	1666	1653
R-squared	0.91	0.85	0.60	0.89	0.91

Note: This table shows the effect of IRA on solar loan outcomes with an indicator for a project finance loan interacted with treatment. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q and EIA Form 860.

Overall, these results show that the IRA disproportionately eased financial constraints for smaller producers and, as a result, increased new capacity investments for these producers relative to larger ones.

5 Conclusion

Demand for electric power is projected to soar in coming years, and questions about what kinds of new generation capacity will be built to meet this demand and who will build it are likely to remain top of mind for policy makers and regulators. These investments will require massive upfront costs and take years to produce revenues, making them highly dependent on access to external financing. In this paper, we show that financial constraints play a key role in driving the ultimate effects of policies targeting the electricity sector. This channel can simultaneously explain why raising the costs of fossil generation caused by MATS reduced solar energy investments, and why changes to renewable energy tax credits in the IRA disproportionately benefited smaller and newer firms.

Our results highlight both challenges and opportunities for policy makers seeking to influence how the growing demand for power will be met. On one hand, the fact that targeted regulations can generate unintended within-firm spillovers to other kinds of generation activity through firms' internal capital markets suggests the need for a more comprehensive understanding of firms' entire generation portfolios when evaluating potential regulatory consequences. At the same time, the fact that relatively modest declines in financing costs can lead to meaningful swings in the composition of investment suggests that financial market interventions can be highly effective in subsidizing new generation capacity.

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Appendix

A Variable Definitions

- Assets: An obligor's reported assets in millions of dollars, from Y-14Q, with values of zero excluded. We first aggregate to a firm-quarter panel by taking a loan volume-weighted average across all loans in the quarter, and then take a simple average across all observed quarters in the year.

- Bond issuer indicator: An indicator equal to one if a firm ever issued a bond in our sample, from Mergent FISD.

- Collateralized: An indicator corresponding to the definition of "asset-backed lending" from Lian and Ma (2021) that equals one if a loan is reported as being secured by any of the following: 1) Real estate, 2) cash and marketable securities, 3) accounts receivable and inventory, or 4) other. These indicators are collapsed to the firm-quarter level using committed loan volume-weighted averages, and then taken as simple averages across all observed quarters in a year.

- Debt: The sum of an obligor's reported short-term debt and long-term debt in millions of dollars, from Y-14Q, with values of zero excluded. We first aggregate to a firm-quarter panel by taking a loan volume-weighted average across all loans in the quarter, and then take a simple average across all observed quarters in the year.

- Fixed assets: An obligor's reported fixed assets in millions of dollars, from Y-14Q, with values of zero excluded. We first aggregate to a firm-quarter panel by taking a loan volume-weighted average across all loans in the quarter, and then take a simple average across all observed quarters in the year.

- Interest rate: The interest rate for an outstanding loan in percentage points, from Y-14Q. This variable is aggregated to the firm-year level by taking a weighted average across all outstanding loans to an obligor TIN in a given year.

- Liabilities: An obligor's reported liabilities in millions of dollars, from Y-14Q, with values of zero excluded. We first aggregate to a firm-quarter panel by taking a loan volume-weighted average across all loans in the quarter, and then take a simple average across all observed quarters in the year.

- Loan size: The total committed exposure for an outstanding loan in millions of dollars, from Y-14Q. This variable is aggregated to the firm-quarter level by taking a sum across all outstanding loans to an obligor TIN and then aggregated to the firm-year level by taking a simple average across all observed firm-quarters in a given year.

- Loss Given Default (LGD): The estimated average loss given default per dollar of loan commitment, from Y-14Q. This variable is aggregated to the firm-quarter level by taking a sum across all outstanding loans to an obligor TIN and then aggregated to the firm-year level by taking a simple average across all observed firm-quarters in a given year.

- Maturity: Remaining average maturity in months, weighted by the committed dollar amount of each loan at the firm-quarter level and then taken as a simple average across all observed quarters in the year, from Y-14Q.

- New loan indicator: An indicator equal to one if a firm receives a new loan, and zero otherwise, from Y-14Q. Data are aggregated to the firm-quarter level using unweighted sums, and then to the firm-year level by taking simple averages across quarters.

- Probability of Default (PD): The averaged expected annual default rate over the life of the loan, weighted by the committed dollar amount of each loan at the firm-quarter level and then calculated as a simple average across all quarters observed in a year, from Y-14Q.

- Project finance indicator: An indicator equal to one of either of the following hold: 1) A loan's obligor is reported by the bank to be a bankruptcy-remote special purpose entity whose primary source of repayment depends on the performance of the specified underlying assets, or 2) a loan's purpose is reported by the bank as "project finance", from Y-14Q. We first take a committed loan volume-weighted average across all loans to the obligor in a given quarter, and then take the simple average across all quarters observed in the year.

- Publicly traded indicator: An indicator equal to one if an obligor reports having publicly traded equity and zero otherwise, from Y-14Q.

- Revolver: An indicator equal to one if a loan is a revolving line of credit, aggregated to the firm-quarter level as an average weighted by committed loan amount and then taken as a simple average across all quarters observed in a year, from Y-14Q.

- Sales: An obligor's reported sales in millions of dollars, from Y-14Q, with values of zero excluded. This variable is aggregated to the firm-quarter level by taking a sum across all outstanding loans to an obligor TIN and then aggregated to the firm-year level by taking a simple average across all observed firm-quarters in a given year.

- Total number of banks: The total number of unique banks lending to a borrower at a time, aggregated to the firm-year level using a simple average across quarters, from Y-14Q.

- Total number of loans: The total number of outstanding loans to a firm across all banks, aggregated to the firm-year level using a simple average across quarters, from Y-14Q.

- Value of bonds issued: The total value of bonds issued by a given obligor in a year in millions of dollars *conditional on issuing at least one bond*, from Mergent FISD.

B Additional Data Description and Policy Background

This section provides additional details on our data sources, policy background, and data construction procedures. We begin by describing bond issuance patterns in the electric power sector, then provide background on tax equity swaps and their role in renewable energy finance before the IRA. We also discuss how MATS compliance varied by regulatory status and describe our procedure for matching Y-14Q loan data to EIA generation data.

B.1 Bond Information

Figure A1 below reports bond issuance statistics for electric power utilities from Mergent FISD. We classify issuance using the reported industry code from Mergent's issuer file. We restrict the sample to issuers with reported four-digit NAICS codes of 2211. The blue and red bars in each figure show issuance for borrowers with reported industry codes for solar or fossil power generation, respectively, from 2012-2014. The green bars show all other generation, and the orange bars show electric power transmission and distribution (T&D). Because Mergent reports only one industry code for each issuer, it is possible that we may be incorrectly classifying some bonds for borrowers with multiple lines of business. Nonetheless, the low volume of bond issuance is consistent with the borrower-level summary statistics reported in Section 2.2.

Figure A1: Bond issuance

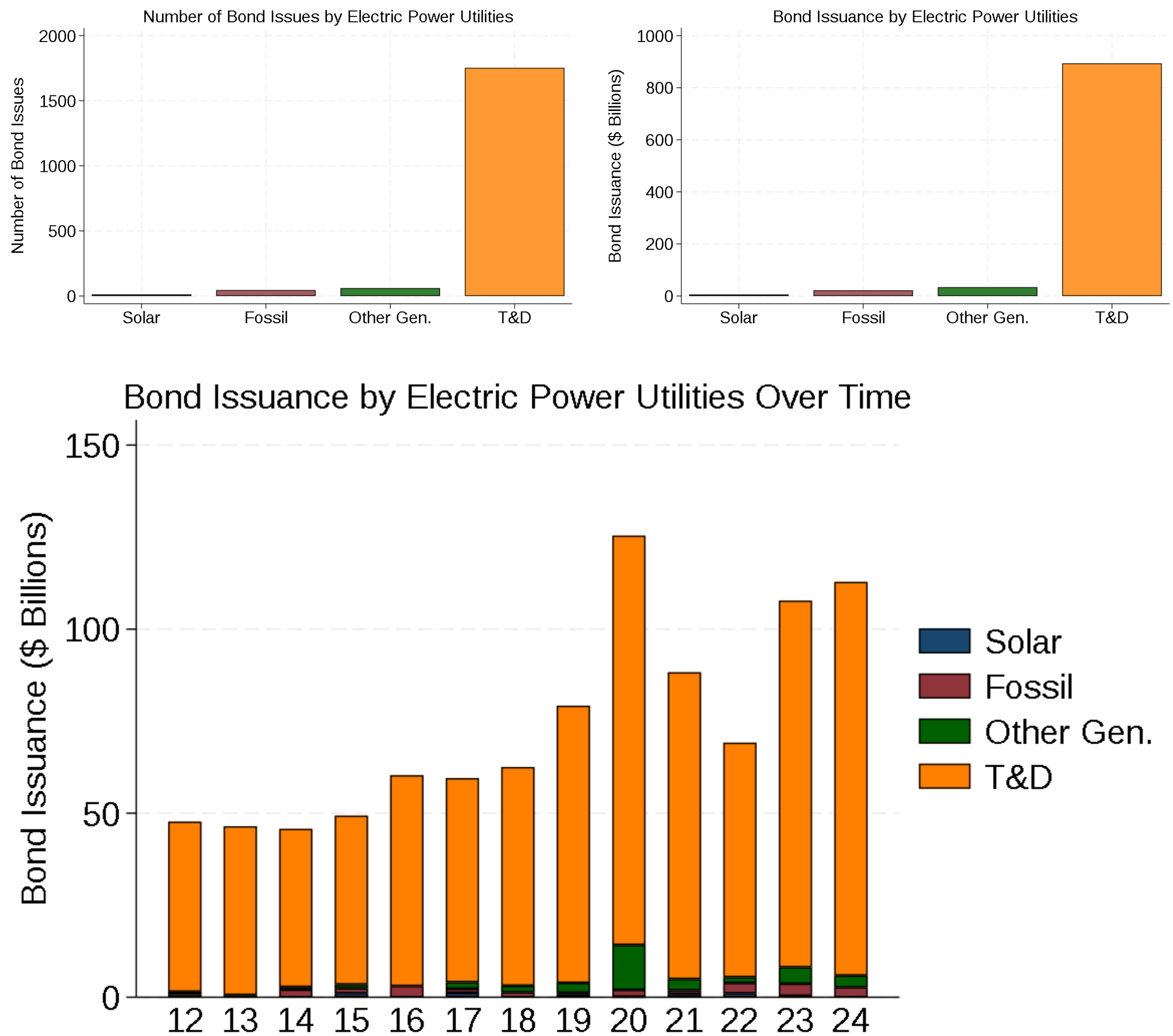


Figure A2: Note: The top panels report the number of bonds (left) or total value of bonds (right) issued by all electric power utilities from 2012-2024. The bottom figure shows a time series of the total value by year. Source: Mergent FISD.

B.2 Tax equity swaps

Smaller solar and wind developers typically do not have the tax liabilities that would allow them to directly take advantage of the investment tax credit (ITC) or the production tax credit (PTC). Before the IRA introduced transferable tax credits, small developers wanting to benefit from the ITC or PTC had to use "tax equity swaps", where an external partner would contribute money to the project in exchange for both a portion of the tax credit and a future stream of payments over the life of the project. Payments to the tax equity investor are defined in terms of after-tax flip yields, which can range from 6.5% to 8% (Cohen, Sternthal, Knapp, and McKenna, 2017). The structure of these deals guaranteed the tax equity investor a specific return on their initial investment.

For example, in the case of the ITC for a solar project, the tax equity investor would receive a large share of the initial ITC payment, and then receive payments from the revenue generated by the project until their after-tax flip yield has been satisfied. At that point, the structure changes ("flips"), with the developer retaining most of the revenues from the solar project. Because the structure of these deals is complicated and project-specific, it is difficult to pin down what share of the ITC is lost to a tax equity swap, but it is likely in the 10% range. Either way, the relatively high after-tax flip yields demanded by tax-equity investors shows that while tax equity swaps allow smaller firms or project finance entities to take advantage of the ITC or the PTC, it meaningfully reduces the value of the pre-IRA subsidy compared to a transferable subsidy.

There are also fixed costs associated with implementing a tax equity swap. Weaver (2024) lists that tax equity investments are typically at least $1 or $2 million to offset the at least $75,000 fees associated with creating the tax equity swap. As a result, renewable projects benefiting from the ITC and PTC were likely to be larger before the transferability provision of the IRA.

One tax advantage remained for larger firms after the transferability of the ITC and PTC: the ability to take advantage of the Modified Accelerated Cost-Recovery System (MACRS). MACRS allows an accelerated depreciation schedule for capital investments for a variety of investments (including non-energy projects), which reduces a firms tax liability (Congressional Budget Office, 2025). Small firms without tax liabilities that directly transfer their ITC or PTC benefits still cannot take advantage of MACRS unless they use a tax equity swap. MACRS was unchanged by the IRA, so it does not affect our results in section 4, but it remains an advantage that larger firms have in financing renewable projects.

B.3 Regulation and MATS

Table A1 provides statistics on coal fired power plants and compliance with MATS by regulatory status. Deregulated coal plants were more likely to have scrubbers in 2014 than regulated coal plants. This may seem counter intuitive based on the Averich Johnson affect outlined in Fowlie (2010), but by 2014, a lot of deregulated coal capacity had retired. Of the remaining deregulated coal, plants with scrubbers were less likely to retire than plants without scrubbers.

Table A1: How coal plants respond to MATS by regulatory status

Action	Deregulated	Restructured
Already in compliance (MW)	67,324	108,054
Install scrubber (MW)	15,306	105,925
Converted to natural gas (MW)	2,128	7,398
Retired (MW)	1,494	18,204
Total coal (MW)	86,251	239,581

Note: The table shows how coal plants active in 2014 responded by the 2016 MATS compliance deadline, broken down by regulatory status. Source: EIA Form 860.

For the plants that were not in compliance with MATS, regulated and deregulated plants responded in similar ways. Around 81% of both regulated and deregulated plants installed scrubbers to comply with MATS. Of the plants that did not installs scrubbers, deregulated plants were a bit more likely to convert to natural gas, while regulated plants were more likely to retire.

B.4 Cleaning and matching of Y-14Q to EIA data

This section describes the cleaning of the Y-14Q data and matching to the EIA-860 data. First, we limit the Y-14Q to NAICS codes related to electric power generation (NAICS

221111-221118). Next, we clean the Y-14Q data. The Y-14Q data provides information on the name of the entity taking out the loan and the tax identification number (TIN). The Y-14Q is loan-level data, and the same name and TIN are not always used across every loan from the same firm. Occasionally, either the name or the TIN is missing from an observation. In cases where either the name or the TIN is missing, we create a new identifier grouping firms that have either the same TIN or name.

Matching is done using the firm name from the Y-14Q and the utility name in the EIA 860 data. We focus on firms that had loans for fossil fuel electricity power generation (NAICS 221112) in the year before treatment (2014 Q2 to 2015 Q2). Matching was prioritized for firms that had larger fossil fuel electric power generation Y-14Q bank loan exposure.

One complication is that data is reported in the EIA 860 data at the utility level, which may not match the firm definition in the Y-14Q data.[21] In many cases, parent companies will own multiple utilities that are reported as separate entities in the EIA 860 data. For example in 2014, American Electric Power owned many utilities including Appalachian Power, Indiana Michigan Power, Kentucky Power, the public service company of Oklahoma, and Southwestern Electric power. All of these utilities are reported in the EIA 860 data under their utility names, not as American Electric power. However, bank loans may have been obtained by any of the utilities, or its parent company, American Electric power, to fund operations. The parent company structure makes it complicated to match the Y-14Q bank loan data to their associated generation portfolios.

To deal with the complicated firm structure, manual review was used to appropriately aggregate utilities and bank loans to the correct parent company. There is no official database linking these firms, so the authors' judgement was used to best aggregate to the firm level. The matching process resulted in a large share of coal generation in the EIA 860 being matched to loans in the Y14. Table A2 shows that 73% of the MW of coal

[21] A utility can be as large as The Tennessee Valley Authority (16 GW of coal generation) or as small as Panther creek power operating LLC (94 MW of coal generation).

recorded by the EIA 860 in 2014 was matched to data in the Y14. The match rate is lower for non-fossil NAICS codes because we prioritized coal generation in the match process.

Table A3 shows that 62% of the total fossil loan volume in the Y-14Q in 2014 was matched to generation data in the EIA 860. Prioritizing coal generation in the match process means that the solar and wind match rates were much lower. However, since the empirical strategy determines treatment status using a firm's coal generation portfolio, a low match rate for non-fossil loans has a limited effect on the results.

Table A2: Share of EIA MW matched to Y-14Q data

Category	Share
Share coal MW matched to Y14	0.73
Share fossil MW matched to Y14	0.61
Share total MW matched to Y14	0.54
Share solar MW matched to Y14	0.25

Note: This table shows the share of generation (in MW) matched between the EIA 860 2014 data and the 2014 Y14. Source: Y-14Q and EIA Form 860.

Table A3: Share of Y-14Q Loan volume matched to EIA data

Category	Total loan volume (Billions)	Loan volume share matched
Fossil loans	52.1	.62
Solar loans	40.8	.29
Wind loans	8.8	.31

Note: Table shows the share of loan volume (in 2014) by category matched between the EIA 860 2014 data and the Y14. Source: Y-14Q and EIA Form 860

C Additional Summary Statistics

This section presents comprehensive summary statistics for all electric power generation sectors covered in our analysis. We report borrower characteristics and loan terms for nuclear, hydro, wind, and other generation types, as well as transmission and distribution utilities. We also provide statistics for loans at origination, comparisons with Compustat data for publicly traded firms, and summary statistics specific to our MATS and IRA analysis samples.

C.1 Additional borrower summary statistics

Table A4: Summary Statistics for Nuclear Borrowers in the Y-14Q Data

	Mean	Median	5%	95%	SD	N
Panel A: All Y-14 borrowers						
Total number of loans	2.09	1.00	1.00	6.00	1.60	431
New loan (indicator)	0.25	0.00	0.00	1.00	0.56	431
Number of banks	4.47	4.00	1.00	13.00	3.43	431
Total loan volume ($ mil)	329.84	118.57	10.80	1,392.16	570.89	431
Interest rate (pp)	2.52	2.25	1.03	4.25	1.17	174
Maturity (months)	76.10	72.00	32.80	124.00	29.12	423
PD (pp)	0.32	0.11	0.03	1.38	0.82	303
LGD (ratio)	0.39	0.39	0.27	0.49	0.09	303
Collateralized (indicator)	0.17	0.00	0.00	1.00	0.35	431
Project finance (indicator)	0.01	0.00	0.00	0.00	0.09	431
Assets ($ mil)	26,265.19	16,105.56	1,700.52	74,620.50	25,314.82	407
Fixed assets ($ mil)	18,392.04	11,852.62	1,616.41	51,208.00	16,955.70	402
Liabilities ($ mil)	18,791.01	11,224.85	1,113.57	53,125.50	17,840.05	406
Debt ($ mil)	8,026.50	5,139.89	664.22	21,544.58	7,651.46	400
Sales ($ mil)	6,845.75	4,280.40	498.74	17,978.00	6,274.89	410
Publicly traded (indicator)	0.64	1.00	0.00	1.00	0.45	431
Bond issuer (indicator)	0.35	0.00	0.00	1.00	0.48	431
Value of bonds issued ($ mil)	1,029.40	700.00	200.00	2,669.76	1,107.93	153
Panel B: All loan characteristics						
Loan size ($ mil)	56.05	27.17	2.30	232.23	73.71	900
Interest rate (pp)	2.58	2.30	1.03	4.50	1.29	298
Maturity (months)	72.64	69.33	21.00	139.00	35.22	842
PD (pp)	0.28	0.11	0.03	0.67	0.92	641
LGD (ratio)	0.39	0.41	0.21	0.49	0.10	640
Term loan (indicator)	0.09	0.00	0.00	1.00	0.28	900
Revolver (indicator)	0.77	1.00	0.00	1.00	0.42	900
Collateralized (indicator)	0.20	0.00	0.00	1.00	0.40	900
Project finance (indicator)	0.01	0.00	0.00	0.00	0.11	900
Assets ($ mil)	30,380.76	18,477.00	1,700.52	85,257.50	27,558.71	821
Fixed assets ($ mil)	20,889.51	13,417.33	1,252.97	57,520.75	18,343.90	816
Liabilities ($ mil)	21,681.16	13,188.60	1,074.00	61,384.25	19,550.27	819
Debt ($ mil)	8,986.50	5,927.75	625.95	23,951.67	8,335.38	812
Sales ($ mil)	7,835.27	4,385.84	51.21	19,624.00	6,917.11	844

Note: Panel A shows summary statistics for the firms in our sample calculated by aggregating information about all loans with a reported NAICS code 221113 (nuclear electricity generation) to the firm-quarter level using obligor TIN. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. All variables come from the Y-14Q except for "Bond issuer" and "Value of bonds issued", which come from Mergent FISD. Panel B shows summary statistics for all loans with a reported NAICS code 221113 (nuclear electricity generation). All variables are calculated as simple averages across all loans. In both panels, firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. All variables are described in detail in Appendix A.
Source: Y-14Q and Mergent FISD.

Table A5: Summary Statistics for Hydro Borrowers in the Y-14Q Data

	Mean	Median	5%	95%	SD	N
Panel A: All Y-14 borrowers						
Total number of loans	2.05	1.00	1.00	5.00	1.73	1,982
New loan (indicator)	0.35	0.00	0.00	2.00	0.76	1,982
Number of banks	4.62	4.00	1.00	12.00	3.39	1,982
Total loan volume ($ mil)	219.00	86.06	3.70	932.45	399.69	1,982
Interest rate (pp)	3.00	2.88	0.82	5.25	1.87	1,022
Maturity (months)	74.10	65.90	12.00	150.00	47.32	1,925
PD (pp)	1.27	0.16	0.07	2.00	7.96	1,229
LGD (ratio)	0.41	0.40	0.20	0.65	0.13	1,231
Collateralized (indicator)	0.24	0.00	0.00	1.00	0.41	1,982
Project finance (indicator)	0.04	0.00	0.00	0.00	0.19	1,982
Assets ($ mil)	15,053.46	6,413.70	12.35	60,886.00	21,751.56	1,706
Fixed assets ($ mil)	10,654.65	4,565.12	4.10	42,736.38	14,998.70	1,693
Liabilities ($ mil)	10,686.50	4,223.64	5.77	44,238.33	15,800.36	1,706
Debt ($ mil)	5,133.20	2,024.39	3.77	21,271.31	7,865.64	1,668
Sales ($ mil)	3,724.05	1,636.22	0.19	15,161.50	5,085.62	1,764
Publicly traded (indicator)	0.43	0.09	0.00	1.00	0.46	1,982
Bond issuer (indicator)	0.20	0.00	0.00	1.00	0.40	1,982
Value of bonds issued ($ mil)	1,020.48	650.00	200.00	3,250.00	1,449.62	389
Panel B: All loan characteristics						
Loan size ($ mil)	38.42	22.50	1.50	136.05	55.28	4,055
Interest rate (pp)	2.88	2.71	0.61	5.25	1.72	1,535
Maturity (months)	72.02	60.00	12.00	146.00	44.44	3,878
PD (pp)	1.18	0.14	0.06	2.00	8.08	2,193
LGD (ratio)	0.42	0.45	0.20	0.65	0.14	2,197
Term loan (indicator)	0.24	0.00	0.00	1.00	0.42	4,055
Revolver (indicator)	0.61	1.00	0.00	1.00	0.48	4,055
Collateralized (indicator)	0.20	0.00	0.00	1.00	0.39	4,055
Project finance (indicator)	0.04	0.00	0.00	0.00	0.19	4,055
Assets ($ mil)	19,071.94	9,398.59	15.97	69,691.25	23,910.21	3,524
Fixed assets ($ mil)	13,443.48	6,788.47	10.36	48,336.75	16,429.60	3,505
Liabilities ($ mil)	13,519.90	6,293.15	9.45	51,092.25	17,299.00	3,527
Debt ($ mil)	6,205.62	2,838.89	6.11	21,758.50	8,259.80	3,472
Sales ($ mil)	4,914.89	2,436.77	0.57	17,274.67	5,877.08	3,645

Note: Panel A shows summary statistics for the firms in our sample calculated by aggregating information about all loans with a reported NAICS code 221111 (hydroelectric power generation) to the firm-quarter level using obligor TIN. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. All variables come from the Y-14Q except for “Bond issuer" and “Value of bonds issued", which come from Mergent FISD. Panel B shows summary statistics for all loans with a reported NAICS code 221111 (hydroelectric power generation). All variables are calculated as simple averages across all loans. In both panels, firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. All variables are described in detail in Appendix A.
Source: Y-14Q and Mergent FISD.

Table A6: Summary Statistics for Wind Borrowers in the Y-14Q Data

	Mean	Median	5%	95%	SD	N
Panel A: All Y-14 borrowers						
Total number of loans	2.17	1.00	1.00	6.00	2.72	772
New loan (indicator)	0.37	0.00	0.00	2.00	0.99	772
Number of banks	3.40	3.00	1.00	8.00	2.44	772
Total loan volume ($ mil)	136.69	75.46	4.60	531.09	190.46	772
Interest rate (pp)	3.44	3.33	1.81	5.42	1.17	551
Maturity (months)	100.52	96.00	11.00	185.80	52.99	771
PD (pp)	2.20	0.51	0.15	11.00	7.48	643
LGD (ratio)	0.37	0.39	0.13	0.60	0.15	643
Collateralized (indicator)	0.40	0.00	0.00	1.00	0.45	772
Project finance (indicator)	0.45	0.23	0.00	1.00	0.47	772
Assets ($ mil)	8,293.50	297.77	16.09	64,437.16	23,937.79	432
Fixed assets ($ mil)	5,191.71	270.08	5.08	41,664.00	14,778.02	410
Liabilities ($ mil)	5,966.88	159.86	3.00	49,079.32	17,386.52	433
Debt ($ mil)	2,705.71	143.18	10.55	24,101.57	7,940.97	355
Sales ($ mil)	1,534.96	18.83	0.00	16,342.99	5,160.56	550
Publicly traded (indicator)	0.16	0.00	0.00	1.00	0.35	772
Bond issuer (indicator)	0.05	0.00	0.00	0.00	0.22	772
Value of bonds issued ($ mil)	1,866.84	1,398.02	225.00	6,081.57	1,904.33	38
Panel B: All loan characteristics						
Loan size ($ mil)	25.14	16.49	1.26	78.00	31.25	1,672
Interest rate (pp)	3.39	3.23	1.76	5.45	1.24	1,000
Maturity (months)	95.37	93.88	11.00	188.00	53.75	1,670
PD (pp)	1.56	0.51	0.14	5.07	5.50	1,332
LGD (ratio)	0.37	0.39	0.13	0.60	0.16	1,332
Term loan (indicator)	0.36	0.00	0.00	1.00	0.48	1,672
Revolver (indicator)	0.27	0.00	0.00	1.00	0.45	1,672
Collateralized (indicator)	0.45	0.00	0.00	1.00	0.49	1,672
Project finance (indicator)	0.47	0.00	0.00	1.00	0.50	1,672
Assets ($ mil)	7,635.87	277.65	22.17	45,618.00	22,046.28	914
Fixed assets ($ mil)	4,552.44	228.35	4.46	25,913.00	12,771.93	881
Liabilities ($ mil)	5,598.40	156.73	5.97	36,282.00	16,282.76	915
Debt ($ mil)	1,938.12	130.97	13.16	11,548.00	6,349.44	788
Sales ($ mil)	1,505.88	15.35	0.00	18,141.00	5,037.16	1,195

Note: Panel A shows summary statistics for the firms in our sample calculated by aggregating information about all loans with a reported NAICS code 221115 (wind electricity generation) to the firm-quarter level using obligor TIN. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. All variables come from the Y-14Q except for “Bond issuer" and “Value of bonds issued", which come from Mergent FISD. Panel B shows summary statistics for all loans with a reported NAICS code 221115 (wind electricity generation). All variables are calculated as simple averages across all loans. In both panels, firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. All variables are described in detail in Appendix A.
Source: Y-14Q and Mergent FISD.

Table A7: Summary Statistics for Other Borrowers in the Y-14Q Data

	Mean	Median	5%	95%	SD	N
Panel A: All Y-14 borrowers						
Total number of loans	2.38	2.00	1.00	7.00	2.45	2,146
New loan (indicator)	0.37	0.00	0.00	2.00	0.91	2,146
Number of banks	4.07	4.00	1.00	11.00	3.27	2,146
Total loan volume ($ mil)	233.76	90.00	3.78	973.22	461.49	2,146
Interest rate (pp)	3.37	3.18	1.15	6.14	1.73	1,163
Maturity (months)	77.28	71.34	16.00	148.00	43.24	2,122
PD (pp)	2.55	0.17	0.04	7.69	11.46	1,786
LGD (ratio)	0.37	0.39	0.10	0.54	0.14	1,786
Collateralized (indicator)	0.28	0.00	0.00	1.00	0.42	2,146
Project finance (indicator)	0.08	0.00	0.00	1.00	0.26	2,146
Assets ($ mil)	18,476.93	5,964.77	16.55	85,134.70	28,755.98	1,832
Fixed assets ($ mil)	10,730.23	3,989.45	5.36	50,261.70	15,922.92	1,797
Liabilities ($ mil)	12,471.66	3,935.99	8.73	53,871.05	19,583.14	1,810
Debt ($ mil)	6,088.20	2,112.41	4.95	26,844.00	9,594.01	1,737
Sales ($ mil)	4,118.13	1,435.90	0.00	16,999.14	5,927.74	1,880
Publicly traded (indicator)	0.50	0.57	0.00	1.00	0.48	2,146
Bond issuer (indicator)	0.18	0.00	0.00	1.00	0.38	2,146
Value of bonds issued ($ mil)	1,252.51	700.00	200.00	3,679.99	2,438.04	382
Panel B: All loan characteristics						
Loan size ($ mil)	40.99	23.38	1.76	134.64	94.82	5,113
Interest rate (pp)	3.49	3.25	1.19	6.78	1.89	2,066
Maturity (months)	75.75	65.75	14.00	145.00	47.60	5,031
PD (pp)	2.66	0.18	0.03	9.10	11.96	4,296
LGD (ratio)	0.37	0.39	0.15	0.52	0.14	4,284
Term loan (indicator)	0.18	0.00	0.00	1.00	0.38	5,113
Revolver (indicator)	0.61	1.00	0.00	1.00	0.49	5,113
Collateralized (indicator)	0.23	0.00	0.00	1.00	0.41	5,113
Project finance (indicator)	0.07	0.00	0.00	1.00	0.25	5,113
Assets ($ mil)	21,386.67	10,476.00	16.47	86,236.00	28,592.12	4,455
Fixed assets ($ mil)	13,894.57	6,801.00	5.73	55,504.33	18,055.63	4,397
Liabilities ($ mil)	15,146.44	6,963.15	7.27	61,407.00	20,508.30	4,432
Debt ($ mil)	7,121.89	3,096.08	4.37	28,462.00	9,683.44	4,256
Sales ($ mil)	5,156.78	2,517.44	0.24	17,666.00	6,396.05	4,532

Note: Panel A shows summary statistics for the firms in our sample calculated by aggregating information about all loans with a reported NAICS code 221118 (other electric power generation) to the firm-quarter level using obligor TIN. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. All variables come from the Y-14Q except for "Bond issuer" and "Value of bonds issued", which come from Mergent FISD. Panel B shows summary statistics for all loans with a reported NAICS code 221118 (other electric power generation). All variables are calculated as simple averages across all loans. In both panels, firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. All variables are described in detail in Appendix A.
Source: Y-14Q and Mergent FISD.

Table A8: Summary Statistics for Transmission and Distribution Borrowers in the Y-14Q Data

	Mean	Median	5%	95%	SD	N
Panel A: All Y-14 borrowers						
Total number of loans	2.50	2.00	1.00	6.00	2.22	2,693
New loan (indicator)	0.36	0.00	0.00	2.00	0.92	2,693
Number of banks	4.68	4.00	1.00	12.00	3.73	2,693
Total loan volume ($ mil)	243.24	113.60	5.38	942.17	430.52	2,693
Interest rate (pp)	3.07	2.90	1.19	5.75	1.43	1,556
Maturity (months)	79.04	68.14	24.39	152.00	50.13	2,654
PD (pp)	1.82	0.23	0.05	6.92	7.50	1,971
LGD (ratio)	0.35	0.37	0.13	0.50	0.13	1,964
Collateralized (indicator)	0.27	0.00	0.00	1.00	0.40	2,693
Project finance (indicator)	0.11	0.00	0.00	1.00	0.29	2,693
Assets ($ mil)	14,483.90	3,498.46	52.49	66,021.84	25,001.73	2,382
Fixed assets ($ mil)	9,102.59	2,620.54	18.01	41,582.67	14,750.34	2,353
Liabilities ($ mil)	10,095.79	2,439.19	27.18	46,446.00	17,419.68	2,367
Debt ($ mil)	5,134.63	1,475.64	20.37	23,229.00	8,560.89	2,310
Sales ($ mil)	3,133.94	734.67	0.00	15,248.00	5,172.65	2,508
Publicly traded (indicator)	0.35	0.00	0.00	1.00	0.45	2,693
Bond issuer (indicator)	0.16	0.00	0.00	1.00	0.36	2,693
Value of bonds issued ($ mil)	1,167.59	700.00	250.00	3,400.00	1,537.43	421
Panel B: All loan characteristics						
Loan size ($ mil)	37.56	23.51	2.00	125.00	76.10	6,728
Interest rate (pp)	2.98	2.75	1.18	5.56	1.50	2,953
Maturity (months)	76.73	61.00	22.00	151.00	44.64	6,555
PD (pp)	1.64	0.22	0.05	5.73	7.23	4,712
LGD (ratio)	0.35	0.38	0.15	0.51	0.13	4,691
Term loan (indicator)	0.11	0.00	0.00	1.00	0.32	6,728
Revolver (indicator)	0.68	1.00	0.00	1.00	0.47	6,728
Collateralized (indicator)	0.24	0.00	0.00	1.00	0.42	6,728
Project finance (indicator)	0.11	0.00	0.00	1.00	0.31	6,728
Assets ($ mil)	15,877.94	4,880.07	85.77	65,188.00	24,910.84	5,862
Fixed assets ($ mil)	10,810.80	3,667.66	38.57	46,134.00	16,198.48	5,807
Liabilities ($ mil)	11,374.68	3,431.88	36.80	47,730.50	17,878.96	5,836
Debt ($ mil)	5,787.05	2,076.00	35.92	24,616.00	8,988.04	5,684
Sales ($ mil)	3,581.77	995.00	0.00	16,171.50	5,549.90	6,267

Note: Panel A shows summary statistics for the firms in our sample calculated by aggregating information about all loans with a reported NAICS code 221121, 221122 (electric power transmission and distribution) to the firm-quarter level using obligor TIN. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. All variables come from the Y-14Q except for “Bond issuer" and “Value of bonds issued", which come from Mergent FISD. Panel B shows summary statistics for all loans with a reported NAICS code 221121, 221122 (electric power transmission and distribution). All variables are calculated as simple averages across all loans. In both panels, firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. All variables are described in detail in Appendix A.
Source: Y-14Q and Mergent FISD.

C.2 Loan summary statistics at origination

Table A9: Fossil loan summary statistics at origination

	Mean	Median	5%	95%	SD	N
Loan size ($ mil)	38.01	24.67	2.14	129.00	44.16	617
Interest rate (pp)	3.20	3.04	1.14	6.10	1.54	271
Maturity (months)	55.58	60.00	11.00	99.00	29.20	612
PD (pp)	0.76	0.43	0.06	2.98	0.99	397
LGD (ratio)	0.36	0.37	0.18	0.53	0.13	389
Term loan (indicator)	0.20	0.00	0.00	1.00	0.40	617
Revolver (indicator)	0.54	1.00	0.00	1.00	0.50	617
Collateralized (indicator)	0.27	0.00	0.00	1.00	0.44	617
Project finance (indicator)	0.13	0.00	0.00	1.00	0.34	617
Assets ($ mil)	17,283.24	6,394.86	67.92	64,439.00	25,387.22	401
Fixed assets ($ mil)	12,316.66	4,461.13	47.16	49,413.00	17,179.57	398
Liabilities ($ mil)	12,606.58	4,710.75	30.47	48,468.00	18,388.09	401
Debt ($ mil)	6,903.64	2,413.57	14.83	34,911.00	10,453.51	372
Sales ($ mil)	3,471.59	985.93	0.00	16,453.00	5,519.21	514

Note: This table shows summary statistics for loan and firm characteristics at the time of origination for the loans in our sample with a reported NAICS code 221112 (fossil electricity generation). Firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Source: Y-14Q.

Table A10: Solar loan summary statistics at origination

	Mean	Median	5%	95%	SD	N
Loan size ($ mil)	33.67	15.26	1.41	107.35	87.95	1,105
Interest rate (pp)	3.53	3.37	1.00	6.25	1.70	649
Maturity (months)	53.00	36.00	5.00	151.00	55.84	1,096
PD (pp)	1.95	0.60	0.14	4.91	8.61	692
LGD (ratio)	0.36	0.35	0.08	0.50	0.16	684
Term loan (indicator)	0.42	0.00	0.00	1.00	0.49	1,105
Revolver (indicator)	0.24	0.00	0.00	1.00	0.43	1,105
Collateralized (indicator)	0.50	0.00	0.00	1.00	0.50	1,105
Project finance (indicator)	0.24	0.00	0.00	1.00	0.43	1,105
Assets ($ mil)	7,691.08	504.24	8.85	41,830.00	18,149.59	324
Fixed assets ($ mil)	4,676.01	320.72	0.53	25,913.00	10,687.82	314
Liabilities ($ mil)	5,949.01	310.00	1.70	30,949.00	13,922.47	324
Debt ($ mil)	2,577.76	126.89	1.24	14,955.00	5,837.81	281
Sales ($ mil)	1,389.57	9.63	0.00	10,577.00	3,835.44	504

Note: This table shows summary statistics for loan and firm characteristics at the time of origination for the loans in our sample with a reported NAICS code 221114 (solar electricity generation). Firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Source: Y-14Q.

Table A11: Nuclear loan summary statistics at origination

	Mean	Median	5%	95%	SD	N
Loan size ($ mil)	59.58	35.00	3.01	234.00	71.58	106
Interest rate (pp)	2.10	1.94	0.70	3.59	1.00	43
Maturity (months)	41.80	36.00	7.00	63.00	25.11	103
PD (pp)	0.14	0.10	0.03	0.44	0.13	61
LGD (ratio)	0.42	0.42	0.20	0.76	0.15	60
Term loan (indicator)	0.21	0.00	0.00	1.00	0.41	106
Revolver (indicator)	0.53	1.00	0.00	1.00	0.50	106
Collateralized (indicator)	0.28	0.00	0.00	1.00	0.45	106
Project finance (indicator)	0.01	0.00	0.00	0.00	0.10	106
Assets ($ mil)	38,911.52	42,257.00	292.63	113,856.00	31,902.55	88
Fixed assets ($ mil)	26,968.83	28,044.36	302.00	68,659.00	20,745.79	87
Liabilities ($ mil)	27,437.24	29,693.00	292.63	72,822.00	21,882.57	88
Debt ($ mil)	11,923.66	9,406.00	285.00	36,199.00	10,007.48	85
Sales ($ mil)	9,723.76	11,099.16	127.72	23,925.00	7,158.54	88

Note: This table shows summary statistics for loan and firm characteristics at the time of origination for the loans in our sample with a reported NAICS code 221113 (nuclear electricity generation). Firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Source: Y-14Q.

Table A12: Hydro loan summary statistics at origination

	Mean	Median	5%	95%	SD	N
Loan size ($ mil)	37.18	20.00	1.25	115.81	74.84	699
Interest rate (pp)	2.92	2.73	0.68	5.25	1.62	323
Maturity (months)	48.96	60.00	6.00	103.00	34.20	690
PD (pp)	1.17	0.20	0.07	2.00	8.00	310
LGD (ratio)	0.42	0.45	0.20	0.65	0.14	312
Term loan (indicator)	0.34	0.00	0.00	1.00	0.47	699
Revolver (indicator)	0.44	0.00	0.00	1.00	0.50	699
Collateralized (indicator)	0.25	0.00	0.00	1.00	0.43	699
Project finance (indicator)	0.06	0.00	0.00	1.00	0.24	699
Assets ($ mil)	17,886.56	7,437.41	13.92	66,417.00	23,280.43	461
Fixed assets ($ mil)	12,843.33	5,350.56	5.20	49,430.00	16,447.69	455
Liabilities ($ mil)	12,774.95	4,895.32	8.73	51,591.00	16,977.34	462
Debt ($ mil)	5,862.20	2,240.40	2.80	21,756.00	8,165.76	455
Sales ($ mil)	4,371.03	1,647.90	0.00	17,120.00	5,874.92	519

Note: This table shows summary statistics for loan and firm characteristics at the time of origination for the loans in our sample with a reported NAICS code 221111 (hydroelectric power generation). Firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Source: Y-14Q.

Table A13: Wind loan summary statistics at origination

	Mean	Median	5%	95%	SD	N
Loan size ($ mil)	37.34	24.42	1.80	109.19	40.33	286
Interest rate (pp)	3.39	3.25	1.86	5.65	1.26	165
Maturity (months)	58.48	58.50	5.00	152.00	49.51	286
PD (pp)	1.04	0.38	0.19	3.73	2.86	219
LGD (ratio)	0.40	0.41	0.15	0.50	0.13	219
Term loan (indicator)	0.44	0.00	0.00	1.00	0.50	286
Revolver (indicator)	0.21	0.00	0.00	1.00	0.41	286
Collateralized (indicator)	0.46	0.00	0.00	1.00	0.50	286
Project finance (indicator)	0.42	0.00	0.00	1.00	0.49	286
Assets ($ mil)	14,582.07	1,366.99	20.27	54,759.20	25,185.08	72
Fixed assets ($ mil)	8,120.79	375.45	0.97	25,913.00	14,272.98	72
Liabilities ($ mil)	10,931.79	826.44	11.53	49,079.32	18,848.22	72
Debt ($ mil)	2,581.24	159.07	10.55	11,548.00	6,368.07	50
Sales ($ mil)	2,287.16	3.75	0.00	18,141.00	5,916.15	133

Note: This table shows summary statistics for loan and firm characteristics at the time of origination for the loans in our sample with a reported NAICS code 221115 (wind electricity generation). Firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Source: Y-14Q.

Table A14: Other loan summary statistics at origination

	Mean	Median	5%	95%	SD	N
Loan size ($ mil)	53.73	20.00	1.33	166.67	211.00	796
Interest rate (pp)	3.20	3.15	1.00	6.50	1.79	403
Maturity (months)	48.76	56.00	8.00	98.00	32.95	789
PD (pp)	1.74	0.36	0.04	4.91	7.58	614
LGD (ratio)	0.37	0.38	0.13	0.52	0.14	611
Term loan (indicator)	0.37	0.00	0.00	1.00	0.48	796
Revolver (indicator)	0.40	0.00	0.00	1.00	0.49	796
Collateralized (indicator)	0.29	0.00	0.00	1.00	0.46	796
Project finance (indicator)	0.05	0.00	0.00	0.00	0.21	796
Assets ($ mil)	19,929.93	8,371.50	4.07	75,215.00	26,564.86	612
Fixed assets ($ mil)	14,036.36	5,977.30	1.31	56,205.00	18,649.48	603
Liabilities ($ mil)	14,513.44	6,138.50	2.38	53,133.00	19,475.73	612
Debt ($ mil)	7,026.20	2,823.73	1.22	26,735.00	9,643.65	566
Sales ($ mil)	4,671.47	2,007.30	0.00	16,773.00	5,913.36	642

Note: This table shows summary statistics for loan and firm characteristics at the time of origination for the loans in our sample with a reported NAICS code 221118 (other electric power generation). Firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Source: Y-14Q.

Table A15: Transmission and Distribution loan summary statistics at origination

	Mean	Median	5%	95%	SD	N
Loan size ($ mil)	48.49	25.00	2.03	157.22	161.07	976
Interest rate (pp)	2.96	2.68	1.06	6.00	1.71	429
Maturity (months)	53.16	60.00	11.00	96.00	29.07	969
PD (pp)	1.20	0.26	0.05	2.88	7.16	601
LGD (ratio)	0.36	0.38	0.20	0.52	0.13	593
Term loan (indicator)	0.20	0.00	0.00	1.00	0.40	976
Revolver (indicator)	0.56	1.00	0.00	1.00	0.50	976
Collateralized (indicator)	0.24	0.00	0.00	1.00	0.43	976
Project finance (indicator)	0.09	0.00	0.00	1.00	0.29	976
Assets ($ mil)	17,671.79	6,380.10	21.22	67,617.40	24,766.21	696
Fixed assets ($ mil)	12,580.85	4,643.53	6.11	50,263.00	17,137.36	688
Liabilities ($ mil)	13,005.10	4,710.75	14.46	48,886.00	18,303.59	688
Debt ($ mil)	6,711.66	2,645.02	4.95	28,630.00	9,605.53	651
Sales ($ mil)	3,541.59	889.26	0.00	15,993.00	5,512.46	831

Note: This table shows summary statistics for loan and firm characteristics at the time of origination for the loans in our sample with a reported NAICS code 221121, 221122 (electric power transmission and distribution). Firm-level income and balance sheet variables (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Source: Y-14Q.

C.3 Loan summary statistics compared to Compustat

Table A16: Fossil firm-level summary statistics including Compustat

	Mean	Median	5%	95%	SD	N
Panel A: Y-14Q borrowers						
Total number of loans	2.45	2.00	1.00	6.00	2.08	761
Number of new loans	0.32	0.00	0.00	2.00	0.70	761
Number of banks	4.84	4.00	1.00	12.00	3.25	761
Total loan volume ($ mil)	310.43	142.86	10.00	1,362.50	507.50	761
Interest rate (pp)	2.48	2.28	1.11	4.32	1.01	315
Maturity (months)	71.92	66.02	34.31	126.51	26.97	739
PD (pp)	0.63	0.14	0.05	2.34	2.24	555
LGD (ratio)	0.40	0.39	0.23	0.50	0.09	555
Collateralized (indicator)	0.08	0.00	0.00	0.71	0.24	761
Project finance (indicator)	0.01	0.00	0.00	0.00	0.09	761
Assets ($ mil)	20,817.20	12,531.00	1,626.19	63,770.25	23,178.30	747
Fixed assets ($ mil)	14,552.40	8,775.92	1,295.09	46,759.86	15,403.34	741
Liabilities ($ mil)	14,969.10	8,604.30	1,030.15	45,588.17	16,798.28	746
Debt ($ mil)	7,528.39	4,009.53	439.60	28,493.31	8,897.38	741
Sales ($ mil)	5,456.46	3,474.27	554.87	17,799.23	5,415.93	750
Panel B: Compustat borrowers						
Total assets ($ mil)	25,100.43	12,002.00	1,697.67	70,923.00	96,352.47	761
Total liabilities ($ mil)	18,636.63	7,880.90	1,076.97	51,184.00	85,224.35	761
Total debt ($ mil)	8,631.45	3,786.00	445.96	25,252.00	22,457.99	761
Total bank debt ($ mil)	633.11	34.54	0.00	3,961.92	1,442.34	734
Bank debt share of total debt (%)	7.41	0.99	0.00	37.94	14.87	734
Employment (thous)	6.68	3.15	0.44	26.06	10.69	710
Capital expenditures ($ mil)	1,393.21	703.80	83.66	5,090.00	1,677.10	761
Bond issuer (indicator)	0.36	0.00	0.00	1.00	0.48	761
Value of bonds issued ($ mil)	931.35	600.00	200.00	3,250.00	1,083.77	273

Note: These tables report summary statistics for the firms in our sample that have loans with a reported NAICS code 221112 (fossil electricity generation) and can be matched to Compustat using obligor TIN. From this set of loans, we aggregate all variables to the firm-quarter level using obligor TIN. Panel A reports firm and loan characteristics from the Y-14Q data. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. Firm-level income and balance sheet variables reported in panel A (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Panel B reports firm-level summary statistics for the same set of firms from Compustat and CapitalIQ. Bond issuance data come from Mergent FISD. All variables are described in detail in appendix A. Source: Y-14Q, Compustat, CapitalIQ, Mergent FISD.

Table A17: Solar firm-level summary statistics including Compustat

	Mean	Median	5%	95%	SD	N
Panel A: Y-14Q borrowers						
Total number of loans	2.18	1.00	1.00	7.00	1.97	398
Number of new loans	0.31	0.00	0.00	2.00	0.80	398
Number of banks	3.27	3.00	1.00	8.00	2.34	398
Total loan volume ($ mil)	188.78	93.68	7.00	725.67	236.40	398
Interest rate (pp)	3.11	2.79	1.09	5.68	2.23	177
Maturity (months)	79.73	63.59	33.00	160.00	40.96	394
PD (pp)	1.61	0.55	0.08	7.74	3.52	189
LGD (ratio)	0.35	0.33	0.18	0.55	0.12	189
Collateralized (indicator)	0.29	0.00	0.00	1.00	0.40	398
Project finance (indicator)	0.11	0.00	0.00	1.00	0.27	398
Assets ($ mil)	17,154.03	10,163.10	63.03	59,267.00	22,027.02	337
Fixed assets ($ mil)	11,362.89	6,787.35	34.70	41,554.00	13,867.43	332
Liabilities ($ mil)	12,201.27	7,025.85	34.01	43,860.00	16,009.77	336
Debt ($ mil)	5,977.83	2,838.00	16.31	21,059.00	7,595.09	335
Sales ($ mil)	4,643.49	2,278.66	1.00	15,995.69	5,620.64	352
Panel B: Compustat borrowers						
Total assets ($ mil)	29,830.31	12,815.50	1,374.01	111,005.00	67,834.59	396
Total liabilities ($ mil)	21,949.49	8,834.83	830.28	72,822.00	59,196.80	396
Total debt ($ mil)	11,304.34	4,962.35	430.70	39,331.00	27,329.74	396
Total bank debt ($ mil)	1,111.83	195.26	0.00	6,105.30	1,858.97	391
Bank debt share of total debt (%)	16.73	5.00	0.00	87.69	25.37	390
Employment (thous)	8.95	4.25	0.12	28.79	17.09	378
Capital expenditures ($ mil)	1,732.22	862.60	76.04	6,020.00	1,996.57	395
Bond issuer (indicator)	0.36	0.00	0.00	1.00	0.48	398
Value of bonds issued ($ mil)	1,083.51	700.00	150.00	3,679.99	1,177.90	143

Note: These tables report summary statistics for the firms in our sample that have loans with a reported NAICS code 221114 (solar electricity generation) and can be matched to Compustat using obligor TIN. From this set of loans, we aggregate all variables to the firm-quarter level using obligor TIN. Panel A reports firm and loan characteristics from the Y-14Q data. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. Firm-level income and balance sheet variables reported in panel A (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Panel B reports firm-level summary statistics for the same set of firms from Compustat and CapitalIQ. Bond issuance data come from Mergent FISD. All variables are described in detail in appendix A. Source: Y-14Q, Compustat, CapitalIQ, Mergent FISD.

Table A18: Nuclear firm-level summary statistics including Compustat

	Mean	Median	5%	95%	SD	N
Panel A: Y-14Q borrowers						
Total number of loans	2.15	1.50	1.00	6.00	1.64	288
Number of new loans	0.24	0.00	0.00	1.00	0.54	288
Number of banks	4.43	4.00	1.00	12.00	3.27	288
Total loan volume ($ mil)	402.64	156.50	23.13	1,442.85	651.32	288
Interest rate (pp)	2.40	2.01	0.94	5.15	1.28	103
Maturity (months)	81.37	76.17	42.00	124.13	25.77	286
PD (pp)	0.19	0.08	0.03	0.66	0.40	202
LGD (ratio)	0.40	0.42	0.30	0.49	0.08	202
Collateralized (indicator)	0.09	0.00	0.00	1.00	0.25	288
Project finance (indicator)	0.00	0.00	0.00	0.00	0.02	288
Assets ($ mil)	28,817.69	17,214.54	3,414.66	75,942.17	25,859.37	284
Fixed assets ($ mil)	20,215.94	13,012.38	2,196.25	53,789.00	17,454.62	284
Liabilities ($ mil)	20,479.09	12,218.00	2,441.92	55,490.75	18,222.31	283
Debt ($ mil)	8,465.91	5,332.38	859.00	22,461.49	7,827.21	284
Sales ($ mil)	7,691.61	4,921.05	967.22	18,467.00	6,373.67	284
Panel B: Compustat borrowers						
Total assets ($ mil)	31,392.99	18,564.12	3,476.80	97,827.00	29,654.88	288
Total liabilities ($ mil)	22,391.17	13,035.00	2,464.40	68,062.00	21,095.41	288
Total debt ($ mil)	10,264.57	6,206.00	859.10	35,081.00	10,359.58	288
Total bank debt ($ mil)	673.15	38.93	0.00	4,113.64	1,474.30	286
Bank debt share of total debt (%)	4.85	0.67	0.00	20.20	10.35	286
Employment (thous)	9.10	6.37	0.60	26.70	8.16	279
Capital expenditures ($ mil)	2,209.07	1,332.06	231.16	6,143.70	2,017.37	288
Bond issuer (indicator)	0.47	0.00	0.00	1.00	0.50	288
Value of bonds issued ($ mil)	986.74	650.00	200.00	2,900.00	1,151.80	134

Note: These tables report summary statistics for the firms in our sample that have loans with a reported NAICS code 221113 (nuclear electricity generation) and can be matched to Compustat using obligor TIN. From this set of loans, we aggregate all variables to the firm-quarter level using obligor TIN. Panel A reports firm and loan characteristics from the Y-14Q data. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. Firm-level income and balance sheet variables reported in panel A (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Panel B reports firm-level summary statistics for the same set of firms from Compustat and CapitalIQ. Bond issuance data come from Mergent FISD. All variables are described in detail in appendix A. Source: Y-14Q, Compustat, CapitalIQ, Mergent FISD.

Table A19: Hydro firm-level summary statistics including Compustat

	Mean	Median	5%	95%	SD	N
Panel A: Y-14Q borrowers						
Total number of loans	2.45	2.00	1.00	6.00	1.90	993
Number of new loans	0.33	0.00	0.00	2.00	0.74	993
Number of banks	5.94	4.00	1.00	14.00	3.84	993
Total loan volume ($ mil)	336.26	163.66	13.91	1,270.75	509.09	993
Interest rate (pp)	2.46	2.08	0.64	4.75	1.99	346
Maturity (months)	77.84	70.23	24.00	131.72	42.58	988
PD (pp)	0.69	0.13	0.07	0.74	6.60	687
LGD (ratio)	0.43	0.43	0.23	0.62	0.11	687
Collateralized (indicator)	0.12	0.00	0.00	1.00	0.29	993
Project finance (indicator)	0.01	0.00	0.00	0.00	0.10	993
Assets ($ mil)	19,928.90	10,791.82	1,640.60	62,185.21	21,894.71	983
Fixed assets ($ mil)	14,032.14	8,043.68	1,137.71	44,998.34	14,931.20	982
Liabilities ($ mil)	14,027.66	7,451.82	1,108.00	44,873.30	15,771.02	983
Debt ($ mil)	6,512.97	3,334.53	434.51	22,035.00	7,768.64	976
Sales ($ mil)	5,102.37	3,067.50	553.46	16,154.79	5,067.77	984
Panel B: Compustat borrowers						
Total assets ($ mil)	24,316.55	11,045.00	1,879.39	78,318.00	40,103.52	991
Total liabilities ($ mil)	17,705.67	7,511.13	1,339.32	56,984.00	33,607.79	991
Total debt ($ mil)	7,755.01	3,612.11	489.95	28,738.00	9,977.66	991
Total bank debt ($ mil)	486.70	25.76	0.00	3,230.42	1,219.58	963
Bank debt share of total debt (%)	6.64	0.66	0.00	31.79	14.65	963
Employment (thous)	6.82	3.39	0.59	23.80	8.00	930
Capital expenditures ($ mil)	1,530.58	755.00	119.05	5,501.00	1,751.48	991
Bond issuer (indicator)	0.36	0.00	0.00	1.00	0.48	993
Value of bonds issued ($ mil)	969.71	600.00	200.00	3,000.00	1,464.32	359

Note: These tables report summary statistics for the firms in our sample that have loans with a reported NAICS code 221111 (hydroelectric power generation) and can be matched to Compustat using obligor TIN. From this set of loans, we aggregate all variables to the firm-quarter level using obligor TIN. Panel A reports firm and loan characteristics from the Y-14Q data. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. Firm-level income and balance sheet variables reported in panel A (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Panel B reports firm-level summary statistics for the same set of firms from Compustat and CapitalIQ. Bond issuance data come from Mergent FISD. All variables are described in detail in appendix A. Source: Y-14Q, Compustat, CapitalIQ, Mergent FISD.

Table A20: Wind firm-level summary statistics including Compustat

	Mean	Median	5%	95%	SD	N
Panel A: Y-14Q borrowers						
Total number of loans	2.37	1.00	1.00	6.00	1.87	67
Number of new loans	0.40	0.00	0.00	3.00	0.94	67
Number of banks	3.09	3.00	1.00	6.00	1.61	67
Total loan volume ($ mil)	268.41	97.09	22.45	1,205.48	366.62	67
Interest rate (pp)	3.30	3.26	1.88	5.74	1.02	51
Maturity (months)	91.14	89.25	43.45	149.16	33.90	67
PD (pp)	0.67	0.58	0.13	1.37	0.46	59
LGD (ratio)	0.37	0.39	0.20	0.50	0.09	59
Collateralized (indicator)	0.38	0.18	0.00	1.00	0.41	67
Project finance (indicator)	0.43	0.30	0.00	1.00	0.43	67
Assets ($ mil)	25,230.22	5,389.86	157.89	117,637.66	36,494.21	56
Fixed assets ($ mil)	14,730.05	4,093.93	152.50	72,180.03	21,037.63	55
Liabilities ($ mil)	18,591.42	3,425.17	34.40	86,097.00	26,954.74	56
Debt ($ mil)	6,270.38	610.43	44.12	33,011.34	10,386.72	46
Sales ($ mil)	6,970.28	332.94	0.00	25,832.00	9,596.78	58
Panel B: Compustat borrowers						
Total assets ($ mil)	85,520.77	34,603.00	2,250.17	127,684.00	242,255.94	67
Total liabilities ($ mil)	67,041.48	28,571.00	1,447.36	90,404.00	215,088.68	67
Total debt ($ mil)	31,628.41	19,464.00	1,393.82	48,091.00	75,477.88	67
Total bank debt ($ mil)	3,232.55	2,167.66	0.00	7,747.53	3,836.01	67
Bank debt share of total debt (%)	19.64	9.20	0.00	68.02	21.35	67
Employment (thous)	19.92	12.77	0.05	34.40	43.65	67
Capital expenditures ($ mil)	3,118.46	2,177.00	0.00	7,624.00	2,963.41	67
Bond issuer (indicator)	0.39	0.00	0.00	1.00	0.49	67
Value of bonds issued ($ mil)	1,841.77	1,200.00	135.00	6,081.57	2,237.14	26

Note: These tables report summary statistics for the firms in our sample that have loans with a reported NAICS code 221115 (wind electricity generation) and can be matched to Compustat using obligor TIN. From this set of loans, we aggregate all variables to the firm-quarter level using obligor TIN. Panel A reports firm and loan characteristics from the Y-14Q data. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. Firm-level income and balance sheet variables reported in panel A (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Panel B reports firm-level summary statistics for the same set of firms from Compustat and CapitalIQ. Bond issuance data come from Mergent FISD. All variables are described in detail in appendix A. Source: Y-14Q, Compustat, CapitalIQ, Mergent FISD.

Table A21: Other firm-level summary statistics including Compustat

	Mean	Median	5%	95%	SD	N
Panel A: Y-14Q borrowers						
Total number of loans	2.83	2.00	1.00	8.00	2.77	969
Number of new loans	0.41	0.00	0.00	2.00	0.96	969
Number of banks	5.04	4.00	1.00	13.00	3.96	969
Total loan volume ($ mil)	353.03	158.90	9.75	1,394.88	614.53	969
Interest rate (pp)	2.77	2.54	0.98	5.14	1.45	427
Maturity (months)	81.86	74.00	32.68	145.11	38.39	967
PD (pp)	0.49	0.12	0.03	1.76	1.58	819
LGD (ratio)	0.39	0.40	0.20	0.52	0.11	819
Collateralized (indicator)	0.17	0.00	0.00	1.00	0.34	969
Project finance (indicator)	0.02	0.00	0.00	0.00	0.15	969
Assets ($ mil)	22,219.30	12,605.72	1,139.16	73,904.09	25,241.53	939
Fixed assets ($ mil)	14,853.72	8,912.03	738.85	51,173.76	16,201.00	928
Liabilities ($ mil)	15,845.20	8,923.97	694.31	52,335.75	18,352.99	939
Debt ($ mil)	7,449.26	4,046.36	313.69	24,983.00	9,021.25	924
Sales ($ mil)	5,624.80	3,194.14	272.00	17,283.67	5,742.41	941
Panel B: Compustat borrowers						
Total assets ($ mil)	41,950.99	11,690.85	1,500.77	83,618.00	219,993.32	968
Total liabilities ($ mil)	33,754.17	7,824.07	1,051.27	62,681.00	197,478.68	968
Total debt ($ mil)	11,400.96	3,777.06	509.54	30,840.00	44,128.09	968
Total bank debt ($ mil)	870.30	47.82	0.00	4,210.43	2,993.96	960
Bank debt share of total debt (%)	9.35	1.60	0.00	49.01	17.32	960
Employment (thous)	8.91	3.50	0.36	27.94	22.34	948
Capital expenditures ($ mil)	1,448.95	725.40	56.00	5,526.00	1,756.58	967
Bond issuer (indicator)	0.34	0.00	0.00	1.00	0.47	969
Value of bonds issued ($ mil)	1,253.13	600.00	200.00	3,600.00	2,737.43	332

Note: These tables report summary statistics for the firms in our sample that have loans with a reported NAICS code 221118 (other electric power generation) and can be matched to Compustat using obligor TIN. From this set of loans, we aggregate all variables to the firm-quarter level using obligor TIN. Panel A reports firm and loan characteristics from the Y-14Q data. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. Firm-level income and balance sheet variables reported in panel A (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Panel B reports firm-level summary statistics for the same set of firms from Compustat and CapitalIQ. Bond issuance data come from Mergent FISD. All variables are described in detail in appendix A. Source: Y-14Q, Compustat, CapitalIQ, Mergent FISD.

Table A22: Transmission and Distribution firm-level summary statistics including Compustat

	Mean	Median	5%	95%	SD	N
Panel A: Y-14Q borrowers						
Total number of loans	2.93	2.00	1.00	8.00	2.51	974
Number of new loans	0.42	0.00	0.00	2.00	0.90	974
Number of banks	5.73	4.00	1.00	13.00	4.22	974
Total loan volume ($ mil)	382.28	175.00	14.61	1,522.43	601.45	974
Interest rate (pp)	2.47	2.29	1.09	4.31	1.07	413
Maturity (months)	72.10	66.24	34.57	128.00	27.57	956
PD (pp)	0.74	0.15	0.05	2.09	4.38	710
LGD (ratio)	0.39	0.40	0.25	0.48	0.08	710
Collateralized (indicator)	0.10	0.00	0.00	0.84	0.25	974
Project finance (indicator)	0.01	0.00	0.00	0.00	0.08	974
Assets ($ mil)	22,365.30	14,291.00	1,663.35	71,526.00	23,285.99	957
Fixed assets ($ mil)	15,597.45	9,976.97	1,252.23	50,731.00	15,735.82	956
Liabilities ($ mil)	16,038.03	10,062.00	1,076.01	52,240.00	16,968.27	955
Debt ($ mil)	7,776.39	4,778.42	461.54	26,068.77	8,444.20	953
Sales ($ mil)	5,356.18	3,582.06	468.77	16,474.00	5,182.97	962
Panel B: Compustat borrowers						
Total assets ($ mil)	24,949.42	12,396.50	1,730.67	70,923.00	85,809.57	974
Total liabilities ($ mil)	18,354.09	8,525.00	1,145.08	53,423.00	75,749.92	974
Total debt ($ mil)	8,460.26	4,225.75	505.95	25,252.00	20,179.81	974
Total bank debt ($ mil)	592.64	35.94	0.00	3,797.77	1,362.76	945
Bank debt share of total debt (%)	7.37	1.12	0.00	35.56	15.31	945
Employment (thous)	6.71	3.42	0.49	23.00	9.89	914
Capital expenditures ($ mil)	1,494.36	800.53	85.40	5,484.00	1,718.90	974
Bond issuer (indicator)	0.37	0.00	0.00	1.00	0.48	974
Value of bonds issued ($ mil)	1,032.19	612.50	225.00	3,250.00	1,514.19	356

Note: These tables report summary statistics for the firms in our sample that have loans with a reported NAICS code 221121, 221122 (electric power transmission and distribution) and can be matched to Compustat using obligor TIN. From this set of loans, we aggregate all variables to the firm-quarter level using obligor TIN. Panel A reports firm and loan characteristics from the Y-14Q data. The total counts of new loans, total loans, unique lenders, and total loan volumes are calculated as sums calculated across all outstanding loans in the quarter, while all other variables are reported as loan-volume weighted averages. Firm-level income and balance sheet variables reported in panel A (assets, fixed assets, liabilities, debt, and sales) are winsorized at the 99th percentile. Panel B reports firm-level summary statistics for the same set of firms from Compustat and CapitalIQ. Bond issuance data come from Mergent FISD. All variables are described in detail in appendix A. Source: Y-14Q, Compustat, CapitalIQ, Mergent FISD.

C.4 MATS and IRA sample characteristics

This subsection presents summary statistics for the specific samples used in our MATS and IRA analyses, showing pre-treatment characteristics of treated and control firms.

Table A23: Summary statistics of firms in IRA sample

	Mean	SD	P10	P50	P90
Solar committed exposure ($mil)	45.42	98.23	1.98	18.80	112.84
Solar interest rate (%)	2.89	1.69	1.12	2.65	5.14
Solar maturity (months)	62.36	49.57	10.00	53.28	135.00
Solar percent of loans asset securitized	52.55	46.90	0.00	60.24	100.00
Solar probability of default (%)	2.59	11.71	0.16	0.60	2.32

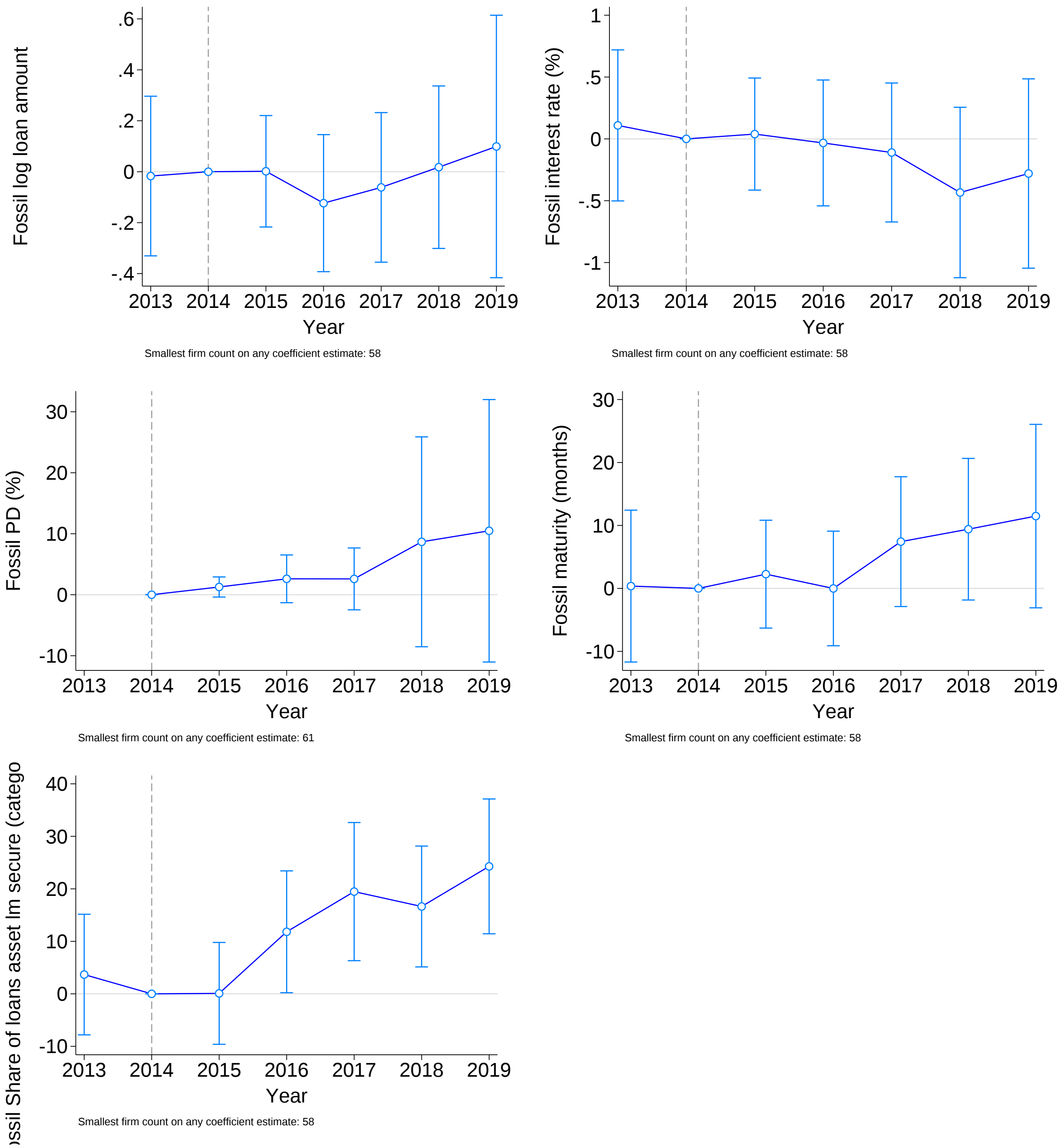
Note: This table shows summary statistics before the IRA went into effect (2019-2021). The smallest category in this table contains 4,000 observations from 391 firms. Source: Y-14Q.

D Additional MATS Results

This section presents additional empirical results from our analysis of the Mercury and Air Toxics Standards (MATS). We provide event study figures showing the dynamic effects of MATS on loan outcomes, robustness checks using alternative treatment thresholds, and additional regression specifications exploring collateralization patterns and risk measures.

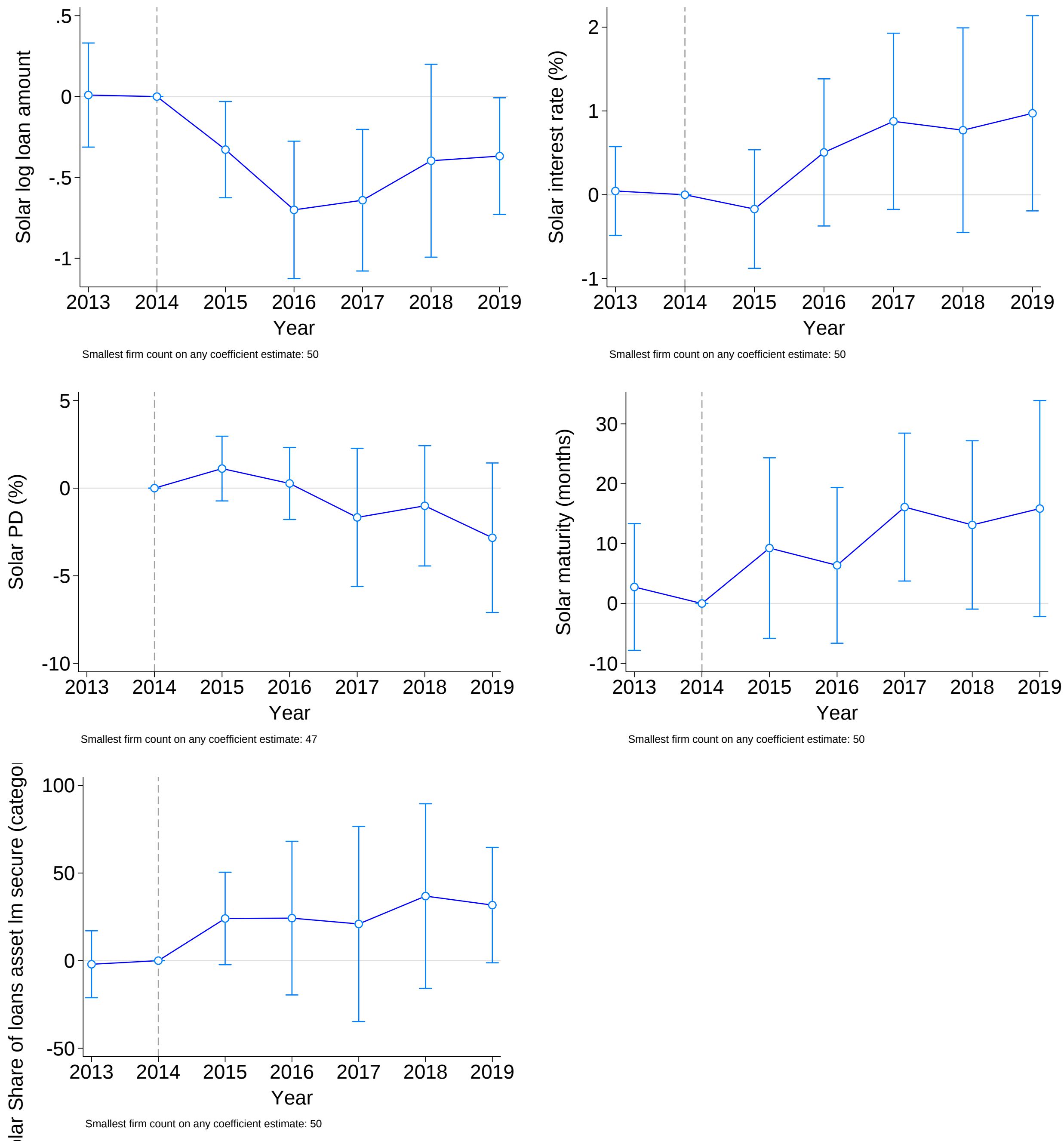
D.1 Event studies

Figure A3: Fossil loan characteristics



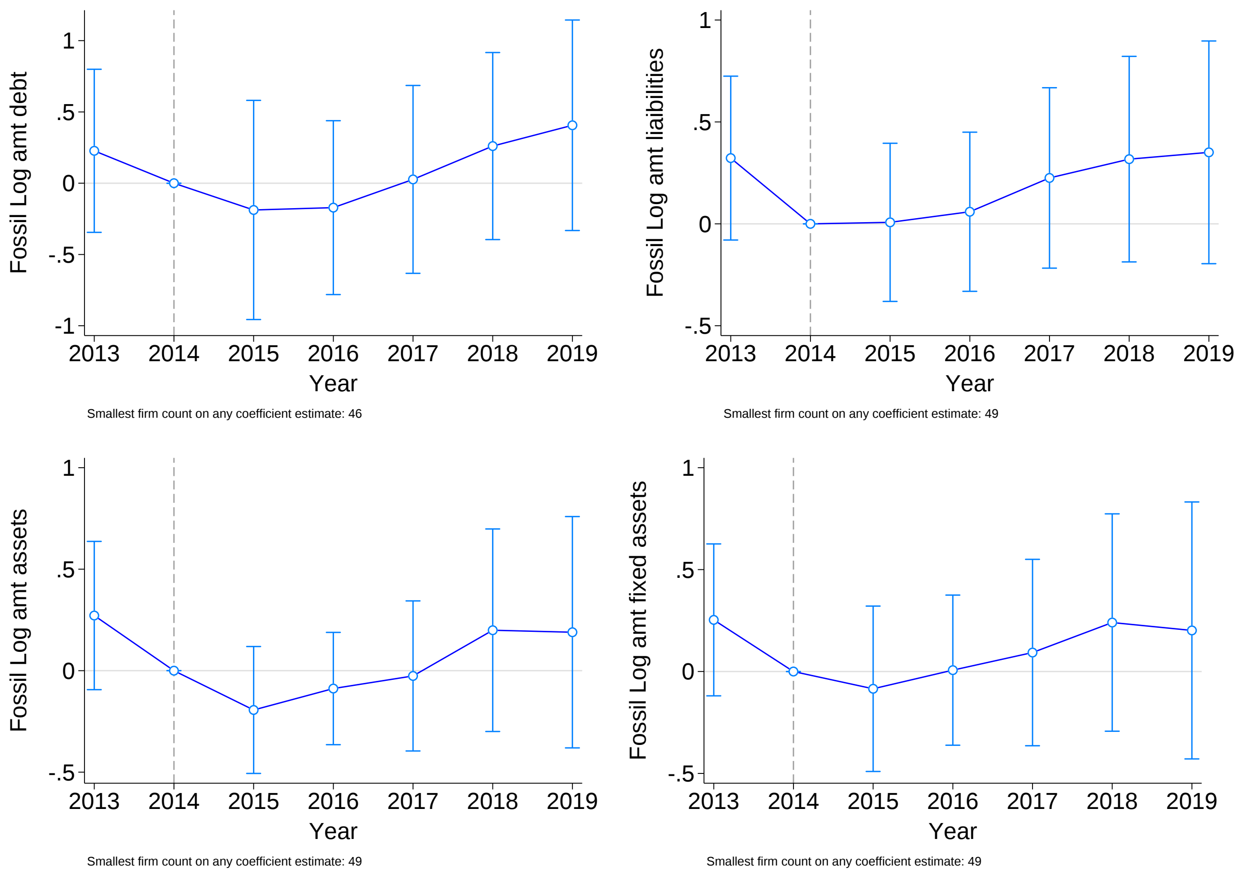
Note: This figure shows the event study estimate from equation 2, focusing on loan outcomes for fossil loans. The Probability of Default variable (PD, middle left panel) is only reported in the Y-14Q data after 2014. Detailed variable definitions are in Appendix A. Error bars show 95% confidence intervals with standard errors clustered at the firm level. Source: Y-14Q and EIA Form 860.

Figure A4: Solar loan characteristics



Note: This figure shows the event study estimate from equation 2, focusing on loan outcomes for solar loans. The Probability of Default variable (PD, middle left panel) is only reported in the Y-14Q data after 2014. Detailed variable definitions are in Appendix A. Error bars show 95% confidence intervals with standard errors clustered at the firm level. Source: Y-14Q and EIA Form 860.

Figure A5: Fossil borrower balance sheet characteristics



Note: This figure shows the event study estimate from equation 2, focusing on loan outcomes for fossil loans. Detailed variable definitions are in Appendix A. Error bars show 95% confidence intervals with standard errors clustered at the firm level.
Source: Y-14Q and EIA Form 860.

D.2 Robustness checks

Table A24: Changes in capacity after MATS with 150 MW treatment cutoff

	Coal (MW)	Solar (MW)	Natural Gas (MW)	Total (MW)
	(1)	(2)	(3)	(4)
post=1 × treatment_150	-823.519^{***}	-417.572^{**}	-49.477	-695.903^{**}
	(215.908)	(200.261)	(461.296)	(348.298)
Firm count	119	119	119	119
Treated firm count	40	29	39	40
Observations	897	897	897	897
R-squared	0.97	0.69	0.99	0.99

Note: Regression estimates from equation 1. This table uses 150 MW as the cutoff for treatment. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: EIA Form 860.

Table A25: Changes in capacity after MATS with 500MW treatment cutoff

	Coal (MW)	Solar (MW)	Natural Gas (MW)	Total (MW)
	(1)	(2)	(3)	(4)
post=1 × treatment_500	-818.332^{***}	-350.666^{*}	37.153	-628.875^{*}
	(230.408)	(181.475)	(417.977)	(343.518)
Firm count	119	119	119	119
Treated firm count	31	21	30	31
Observations	897	897	897	897
R-squared	0.97	0.68	0.99	0.99

Note: Regression estimates from equation 1. This table uses 500 MW as the cutoff for treatment. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: EIA Form 860.

Table A26: Fossil loan outcomes excluding unmatched fossil

	Total loan volume (%)	Interest rate (pp)	Probability of default (pp)	Collateralized (pp)	Maturity (months)
	(1)	(2)	(3)	(4)	(5)
Treatment × Post	0.052	-0.268	5.893	14.792^{**}	6.720
	(0.169)	(0.201)	(5.203)	(6.497)	(5.006)
Firm count	29	29	29	29	29
Observations	133	132	131	133	132
R-squared	0.89	0.84	0.49	0.91	0.81

Note: Regression estimates from equation 1, focusing on fossil loans. This table is similar to table 6, but all the unmatched fossil firms are dropped from the estimation sample. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q and EIA Form 860.

Table A27: Solar loan outcomes without unmatched fossil

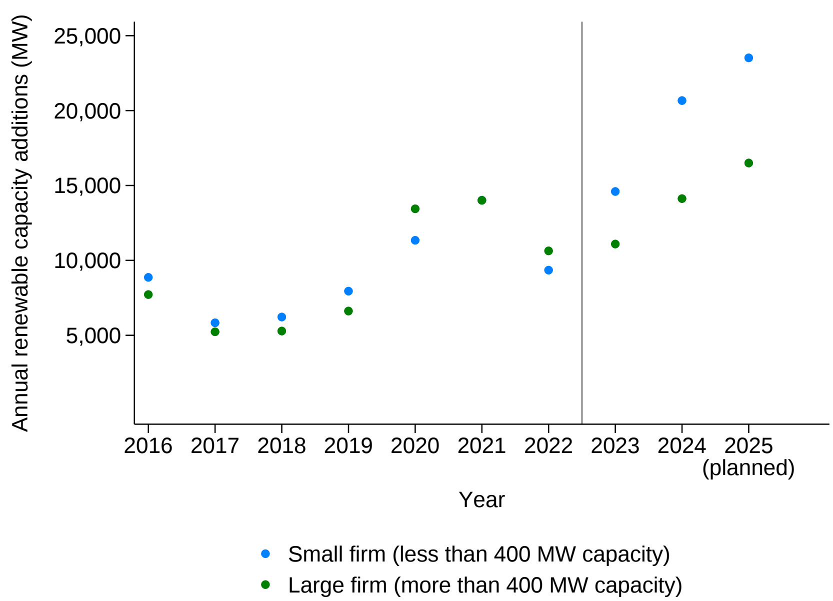
	Total loan volume (%)	Interest rate (pp)	Probability of default (pp)	Collateralized (pp)	Maturity (months)
	(1)	(2)	(3)	(4)	(5)
Treatment × Post	-0.315^{*}	0.867^{*}	-0.761	11.520	9.819^{**}
	(0.178)	(0.461)	(1.171)	(15.425)	(4.759)
Firm count	82	82	80	82	82
Observations	332	332	325	332	332
R-squared	0.92	0.57	0.52	0.83	0.86

Note: Regression estimates from equation 1, focusing on solar loans. This table is similar to table 6, but all the unmatched fossil firms are dropped from the estimation sample. Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q and EIA Form 860.

E Additional IRA Results

This section presents additional results from our analysis of the Inflation Reduction Act (IRA). We show how new solar capacity additions varied by firm size before and after the IRA, including planned capacity through 2025. We also examine how the effects of the IRA differed for project finance loans compared to corporate balance sheet loans.

Figure A6: New renewables capacity added by firm size (including planned)



Note: This figure shows new capacity by year based on maximum firm size (in MW) over the sample. It is the same as figure 7 but it also includes planned 2025 capacity. Source: EIA Form 860.

Table A28: Solar loan outcomes for non-project finance loans

	Total loan volume (%)	Interest rate (pp)	Probability of default (pp)	Collateralized (pp)	Maturity (months)
	(1)	(2)	(3)	(4)	(5)
Treatment × Post	−0.056	−0.346	1.055	−0.322	−7.181**
	(0.074)	(0.266)	(1.364)	(2.494)	(3.246)
Firm count	222	199	296	332	219
Observations	980	858	1384	1589	969
R-squared	0.95	0.85	0.61	0.91	0.87

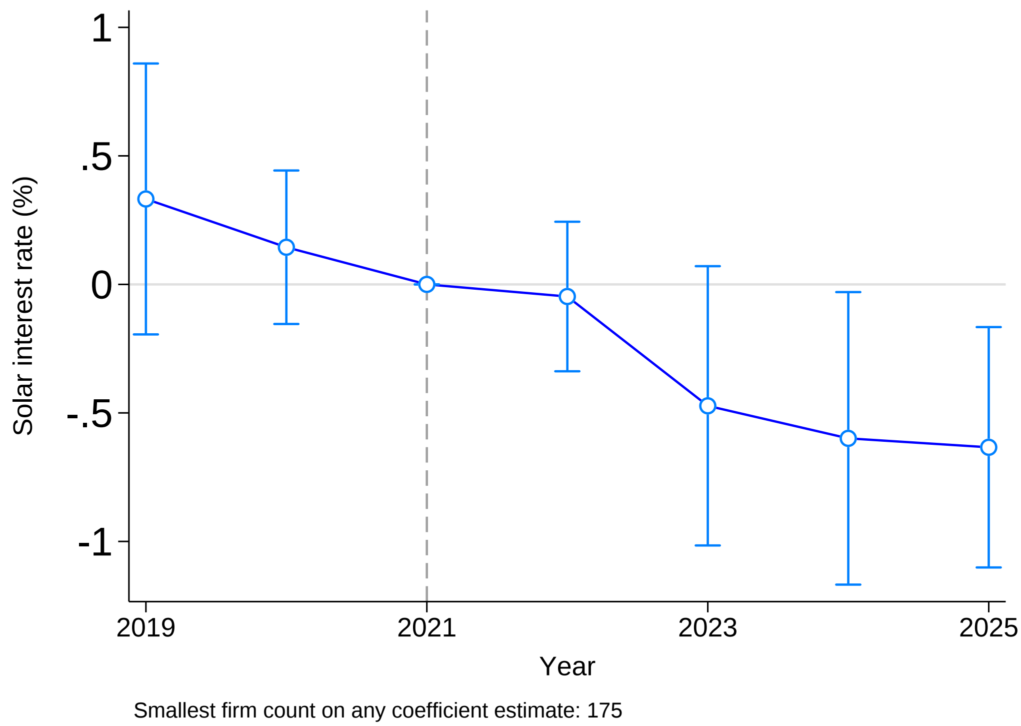
Note: Regression estimates from equation 1, focusing on solar loans where the special purpose indicator is 0 Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q.

Table A29: Solar loan outcomes for project finance loans

	Total loan volume (%)	Interest rate (pp)	Probability of default (pp)	Collateralized (pp)	Maturity (months)
	(1)	(2)	(3)	(4)	(5)
Treatment × Post	0.533***	−1.395***	1.055	−0.322	7.258
	(0.152)	(0.423)	(1.364)	(2.494)	(5.057)
Firm count	71	64	296	332	71
Observations	326	288	1384	1589	326
R-squared	0.92	0.82	0.61	0.91	0.97

Note: Regression estimates from equation 1, focusing on solar loans where the special purpose indicator is 0 Detailed variable definitions are in Appendix A. Standard errors clustered at the firm level. Signif. codes: ***: 0.01, **: 0.05, *: 0.1. Source: Y-14Q.

Figure A7: The effect of the IRA on solar interest rates for small firms



Note: This figure shows the event study estimate from equation 2, applied to the IRA treatment. Detailed variable definitions are in Appendix A. Error bars show 95% confidence intervals with standard errors clustered at the firm level. Source: Y-14Q and EIA Form 860.